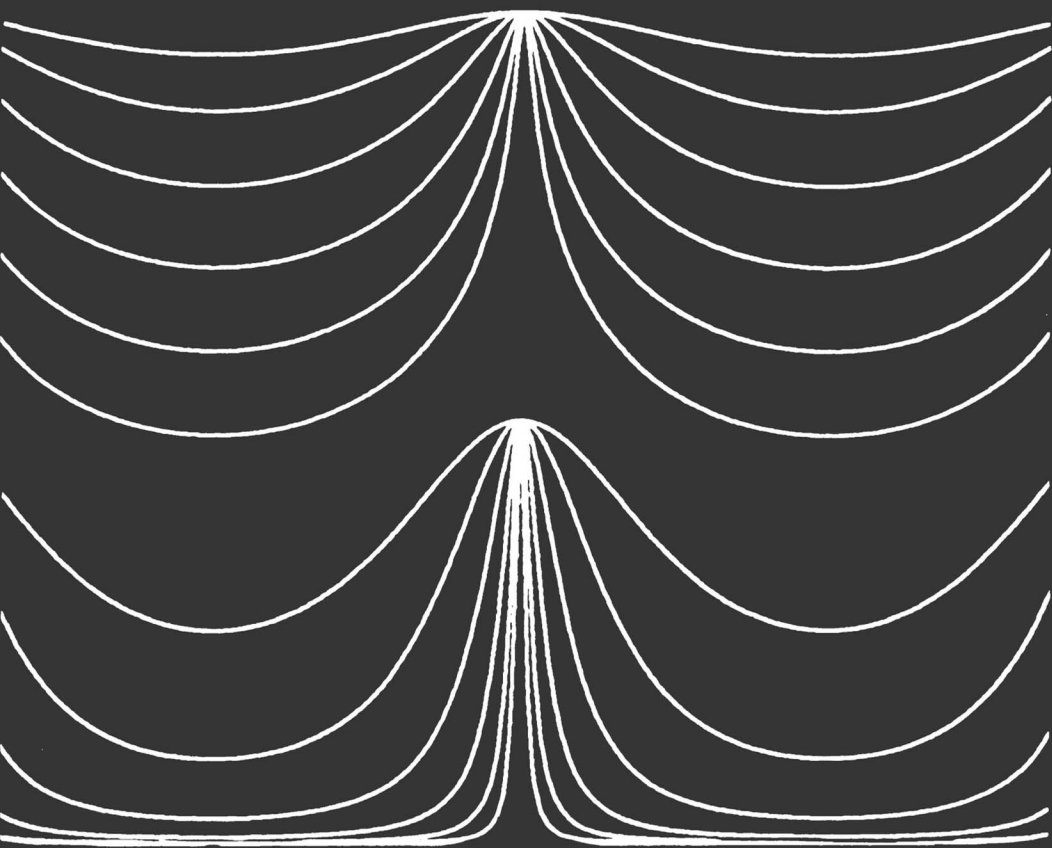

**THE FABRY-PEROT
INTERFEROMETER**

HISTORY, THEORY, PRACTICE
AND APPLICATIONS

J M VAUGHAN



The Fabry–Perot Interferometer

History, Theory, Practice and Applications

Series on Optics and Optoelectronics

Series Editors: **E R Pike** FRS and **W T Welford** FRS

Other books in the series

Aberrations of Optical Systems

W T Welford

Laser Damage in Optical Materials

R M Wood

Waves in Focal Regions

J J Stamnes

Laser Analytical Spectrochemistry

edited by V S Letokhov

Laser Picosecond Spectroscopy and Photochemistry of Biomolecules

edited by V S Letokhov

Cutting and Polishing Optical and Electronic Materials

G W Fynn and W J A Powell

Prism and Lens Making

F Twyman

The Optical Constants of Bulk Materials and Films

L Ward

Infrared Optical Fibers

T Katsuyama and H Matsumura

Solar Cells and Optics for Photovoltaic Concentration

A Luque

Other titles of related interest

Infrared Optical Materials and their Antireflection Coatings

J A Savage

Principles of Optical Disc Systems

G Bouwhuis, J Braat, A Huijser, J Pasman, G van Rosmalen and K Schouhamer
Immink

Thin-film Optical Filters

H A Macleod

Aberration and Optical Design Theory

G G Slyusarev

Lasers in Applied and Fundamental Research

S Stenholm

Series on Optics and Optoelectronics

The Fabry–Perot Interferometer

History, Theory, Practice and Applications

J M Vaughan MA, DPhil

*Royal Signals and Radar Establishment,
Great Malvern, UK*



Taylor & Francis

Taylor & Francis Group

New York London

Published in 1989 by
Taylor & Francis Group
270 Madison Avenue
New York, NY 10016

Published in Great Britain by
Taylor & Francis Group
2 Park Square
Milton Park, Abingdon
Oxon OX14 4RN

© 1989 by Taylor & Francis Group, LLC

No claim to original U.S. Government works

10 9 8 7 6 5 4 3 2

International Standard Book Number-10: 0-85274-138-3 (Hardcover)

International Standard Book Number-13: 978-0-85274-138-2 (Hardcover)

This book contains information obtained from authentic and highly regarded sources. Reprinted material is quoted with permission, and sources are indicated. A wide variety of references are listed. Reasonable efforts have been made to publish reliable data and information, but the author and the publisher cannot assume responsibility for the validity of all materials or for the consequences of their use.

No part of this book may be reprinted, reproduced, transmitted, or utilized in any form by any electronic, mechanical, or other means, now known or hereafter invented, including photocopying, microfilming, and recording, or in any information storage or retrieval system, without written permission from the publishers.

Trademark Notice: Product or corporate names may be trademarks or registered trademarks, and are used only for identification and explanation without intent to infringe.

Library of Congress Cataloging-in-Publication Data

Catalog record is available from the Library of Congress

informa

Taylor & Francis Group
is the Academic Division of Informa plc.

Visit the Taylor & Francis Web site at
<http://www.taylorandfrancis.com>

For Morris and Elsie,
For Charlotte and Judy, and
Sarah, Matthew, Thomas and Sophie.

‘Loving and beloved . . .’

Charles Fabry 1867–1945



Reproduced by permission of *Archives de l'Académie des Sciences de Paris*.

Marie-Paul-Auguste-Charles Fabry was born at Marseille in 1867 into a scientific family. He entered the Ecole Polytechnique in 1885 and became agrégé de physique in 1889; he received his doctorate in 1892 with a thesis entitled 'Théorie de la visibilité et de l'orientation des franges d'interférence'. After teaching assignments at various lycées he returned to Marseille in 1894 as Maître de Conférences at the University, joining the laboratory of Macé de Lépinay where optics was a prime interest. Thus began his long association with Alfred Perot and later with Henri Buisson. From 1904 he was Professor at Marseille and made many interferometric investigations including astronomical studies, his interest in astronomy having developed at an early age in company with his elder brothers. In 1921 he became Professor of Physics at the Sorbonne where he stayed until his retirement in 1937. From its foundation in 1920 he had been organizer and first Director of the Institut d'Optique; from 1927 he also held a Chair of Physics at the Ecole Polytechnique. Charles Fabry thus spent the latter part of his life in Paris where he was elected member of the Académie des Sciences in 1927. He was awarded the Rumford Medal of the Royal Society in 1918, the Draper Medal of the National Academy in 1919 and the Franklin Medal in 1931, the year in which he was elected a foreign member of the Royal Society. During the Great War he had been involved with optics and artillery and in 1917 he took part in an important scientific mission to the United States. Of this visit de Broglie wrote: 'le charme de ses relations personnelles accrut son autorité'. In a life dedicated to science he shared his great enthusiasm for optics in several popular articles and expositions. His latter years, clouded by the invasion of France, were spent on the coast of Provence. After the Liberation he returned to Paris, dying on 11th December 1945.

Drawn mostly from Louis de Broglie, *Obituary Notices of Fellows of the Royal Society* 1947 5 445–50.

Alfred Perot 1863–1925



Reprinted courtesy of *The Astrophysical Journal*, published by the University of Chicago Press; © (1926) The American Astronomical Society.

Jean-Baptiste Gaspard Gustav Alfred Perot was born at Nancy in 1863. In 1882 he entered the Ecole Polytechnique in Paris; on completing his course there in 1884 he returned to Nancy to work in the physical laboratory of the University. He received his Doctor's degree in 1888, his thesis being entitled 'Sur la mesure des volumes spécifiques des vapeurs saturantes, et mesure de l'équivalent mécanique de la chaleur'; in that year he was appointed Maître de Conférences in the Science Faculty at the University of Marseille. In 1894 he became Professor of Industrial Electricity at the University, his successor as Maître de Conférences being Charles Fabry; their long collaboration began at this time. In 1901 Perot became the first director of the Laboratoire d'Essais, newly created in Paris by the French government. After seven years he relinquished this position in order to devote himself to pure science and teaching; he became professor in the Ecole Polytechnique, and physicist at the Observatory of Meudon. During the Great War Perot was particularly engaged in problems relating to telegraphy and communication. It seems that Perot was personally not widely known abroad; he had travelled but little, not willingly leaving his large family. Charles Fabry wrote that 'only those who knew him intimately could appreciate the vivacity and breadth of his spirit'. His career reveals the breadth and versatility of his scientific interests that ranged from expert knowledge of the electrical industry, to his attempts to verify the gravitational red shift of Einstein's general theory of relativity. Most notably, to the instrument that bears his name, he brought great mechanical skill and insight that admirably complemented the optical expertise of his colleague Charles Fabry. Alfred Perot died on November 28th 1925 aged only sixty-two years, survived by five children and many grandchildren.

Drawn mostly from the obituary notice by Charles Fabry, *Astrophysical Journal* 1926 **64** 209–14.

Contents

Series Editors' Preface	xiii
Preface	xv
Acknowledgments	ix
1 Historical Background	1
1.1 Introduction	1
1.2 The Airy distribution	2
1.3 The early work of Fabry and Perot	6
1.4 The papers of 1899	8
1.5 The work of 1900–1902	14
1.6 Progress after 1902	19
1.7 Spectroscopic standards	24
1.8 Application to the study of nebulae	30
1.9 Line broadening, Doppler and gravitational shift, and other studies	32
1.10 Conclusion	36
Translation of passages from Chapter 1	38
Historical references to 1927 in chronological order	40
2 Basic Elements of Optical Spectroscopy	44
2.1 Introduction	44
2.2 Basic spectroscopic information	45
2.3 Spectroscopic terms	48
2.4 Techniques of light detection and recording	53
2.5 Photon statistics and the radiation field	64
2.6 Reflecting films	70
2.7 Cavity modes and resonators	78
3 The Plane Fabry–Perot Interferometer	89
3.1 Introduction	89
3.2 General form of the interference fringes	89
3.3 Line width, finesse and resolving power	104
	ix

3.4	The Fabry–Perot spectrometer	108
3.5	Photographic recording and analysis of fringes	114
3.6	Plate defects	123
3.7	Profile analysis of experimental spectra	127
3.8	Summary of useful expressions	132
4	Practical Operation of Plane Fabry–Perot Interferometers	135
4.1	Introduction	135
4.2	Fixed or static arrangements	138
4.3	Mechanical scanning	157
4.4	Selection and adjustment of optical components	177
5	The Spherical Fabry–Perot Interferometer	184
5.1	Introduction	184
5.2	The confocal interferometer	185
5.3	Defocused and other spherical interferometers	196
5.4	Coherent illumination and mode analysis	202
5.5	Construction and operation of spherical interferometers	207
5.6	Summary of useful expressions	212
6	Multiple and Multi-pass Interferometers	214
6.1	Introduction	214
6.2	Interferometry with multiple étalons for increased spectral range	216
6.3	Multi-pass interferometry for increased contrast	235
7	Applications to Atomic Spectroscopy	254
7.1	Introduction	254
7.2	Measurement of wavelength	254
7.3	Line structure in atomic spectra	260
7.4	Measurement of line broadening and related shift	273
7.5	Radiation correction effects—the Lamb shift	282
7.6	Contributions to laser spectroscopy	286
8	Applications to Astronomy and Astrophysics	296
8.1	Introduction	296
8.2	Extra-terrestrial measurements—imaging	299
8.3	Extra-terrestrial measurements—studies of individual lines	304
8.4	Extra-terrestrial measurements—wide-band and absorption line spectroscopy	309
8.5	Extra-terrestrial measurements—simultaneous high spatial and spectral resolution	320
8.6	Measurements of the Earth’s atmosphere	327
9	Applications to Light Scattering Studies of Materials	338
9.1	Introduction	338

9.2	Basic terms and considerations	339
9.3	Experimental techniques	341
9.4	Experimental review	350
10	Applications to Metrology, Optical Bistability, Velocimetry, Infrared, Sensors, Plasma Physics and Miscellaneous Devices	382
10.1	Introduction	382
10.2	Metrology	383
10.3	Optical bistability	392
10.4	Laser Doppler velocimetry	401
10.5	Operation and applications in the infrared and longer wavelengths	405
10.6	Fabry–Perot interferometers as sensors	422
10.7	Plasma diagnostics	434
10.8	Miscellaneous devices	438
Appendices		447
1	The first Fabry–Perot interferometer	447
2	Measurement of absolute sensitivity of photomultipliers and other experimental considerations	456
3	Fizeau and multiple-beam wedge fringes	460
4	The effect of finite aperture size (ray treatment)	464
5	Multiple and equivalent mirror calculations—the ‘two effective interfaces’ method	468
6	Fourier formulation of the instrumental function and its convolution with Gaussian and Lorentzian functions	471
7	Intra-cavity effects: light fields, sources and absorbers, dispersion	474
8	Evaluation of resolving power and étendue for a plane Fabry–Perot spectrometer—the luminosity–resolution product method	481
9	Voigt functions: properties and simple analysis	485
10	Fundamental limits to the precision of spectroscopic measurements—least uncertainty methods	488
11	Construction and testing of spacers for Fabry–Perot étalons	496
12	Piezoelectric materials and transducers	499
13	Operation of the deformable parallelogram	501
14	Computer control and analysis	504
15	A control system for laser scattering spectroscopy	507
16	Interferometers for single-frequency operation of gas lasers	510
17	Modes in Cartesian and cylindrical coordinates	513
18	Gaussian beam propagation	515
19	Use of retroreflecting prisms in multi-pass operation	517
20	Channel spectra for continuum and absorption line studies	520

xii *The Fabry–Perot Interferometer*

References and Further Reading **523**

Index **579**

Series Editors' Preface

Optics has been a major field of pure and applied physics since the mid 1960s. Lasers have transformed the work of, for example, spectroscopists, metrologists, communication engineers and instrument designers in addition to leading to many detailed developments in the quantum theory of light. Computers have revolutionized the subject of optical design and at the same time new requirements such as laser scanners, very large telescopes and diffractive optical systems have stimulated developments in aberration theory. The increasing use of what were previously not very familiar regions of the spectrum, e.g. the thermal infrared band, has led to the development of new optical materials as well as new optical designs. New detectors have led to better methods of extracting the information from the available signals. These are only some of the reasons for having an *Adam Hilger Series on Optics and Optoelectronics*.

The name Adam Hilger, in fact, is that of one of the most famous precision optical instrument companies in the UK; the company existed as a separate entity until the mid 1940s. As an optical instrument firm Adam Hilger had always published books on optics, perhaps the most notable being Frank Twyman's *Prism and Lens Making*.

Since the purchase of the book publishing company by The Institute of Physics in 1976 their list has been expanded into all areas of physics and related subjects. Books on optics and quantum optics have continued to comprise a significant part of Adam Hilger's output, however, and the present series has some twenty titles in print or to be published shortly. These constitute an essential library for all who work in the optical field.

Preface

Any scientist with even the most cursory knowledge of optics and spectroscopy must be aware of the long and distinguished history of the interferometer first described in detail by Charles Fabry and Alfred Perot of the University of Marseilles. It is perhaps the versatility of such an essentially simple instrument that is most impressive, coupled with the fact that at so many turning points of physics the Fabry–Perot interferometer has provided seminal information. This extends from the earliest days of atomic physics to the present age of coherent light and the laser and includes for example measurements of fine and hyperfine structure in atomic spectra, precision measurements of size and of the velocity of light, contributions to quantum electrodynamics with the earliest observations of the Lamb–Retherford shift, studies of isotope shift and nuclear structure, cavity mode and laser theory, and much recent work in light scattering spectroscopy, astrophysical applications, remote sensing, optical bistability and plasma and atomic physics.

Faced with such a catalogue and aware of the range of variations of the instruments that have now been developed, any author must approach such a subject with considerable trepidation. In this book I have attempted, ambitiously, to provide at once a brief history and an account of theory, practice and applications of the Fabry–Perot interferometer in its various forms. Part of my excuse for embarking on this task was that no such work, devoted exclusively to Fabry–Perot interferometry, had previously been published. Having said this I am keenly aware of my indebtedness to the works of many authors. These include the early monograph of Charles Fabry published in 1923, authoritative reviews by Derek A Jackson, H G Kuhn and P Jacquinot and several monumental papers from the French groups, particularly by R Chabbal, H Chantrel and C Dufour. In addition there are many books, which touch on interferometry to a greater or lesser degree, that I have consulted extensively, including those by M Born and E Wolf, W H Steel, S Tolansky, J F James and R S Sternberg, M Francon, A McLeod and W E Williams to name but a few.

It is perhaps appropriate to outline briefly my own involvement with Fabry–Perot interferometry. This began in 1958 on the laboratory teaching course at the Clarendon Laboratory, Oxford with a student experiment to measure the fine structure constant in $H\alpha$. The aesthetic pleasure of this beautiful experiment and the intellectual excitement in the realisation that such a simple

equipment could explore atomic structure so profoundly and yet so tangibly has never left me. In 1959 I was fortunate enough to start graduate work in the research group of Dr H G Kuhn and for several years used plane Fabry–Perot interferometers, mostly with pressure scanning, in some of the first studies of emission line broadening in atomic spectra. From 1966 to 1968, as a Lecturer in the Palmer Physical Laboratory at Princeton University, I worked with my good friend the late Professor Tom Carver using interference techniques in an attempt to examine rotating superfluid helium. This led to the study at Birmingham University of light scattering from liquid and superfluid helium in many different regimes. For these investigations I developed plane and confocal piezoelectrically scanned interferometers of large spacing and resolving limit approaching 1 MHz. It is a great pleasure to acknowledge the close collaboration with my colleagues Professor W F Viven of Birmingham University, and Dr E R Pike of the Royal Signals and Radar Establishment, during this period. I have since built various confocal and multi-pass interferometers and used them in many light scattering studies, including measurement on biological materials in collaboration with the late Professor Sir John Randall. Much of my work in recent years at the Royal Signals and Radar Establishment has employed the new spectroscopic techniques of photon correlation and laser heterodyning. These have often provided valuable perspectives on interferometry itself — of considerable help in preparing the present work.

The layout of this book is straightforward: the first chapter provides a historical account of Fabry–Perot interferometry, from the earliest years, up to and including the first quarter of the twentieth century. Anyone studying these early papers cannot but be impressed both by the experimental skill and by the foresight they often reveal of the potential range of applications. The second chapter outlines a number of basic optical and spectroscopic concepts and techniques. The third chapter describes the theory of the plane interferometer in its various manners of use, while the fourth chapter is devoted to their instrumental realization. The fifth chapter deals with the theory and practice of confocal and spherical interferometers while the sixth discusses multiple and multi-pass interferometers. These four chapters, three to six, thus form the instrumental core of the book. The final four chapters then deal successively with the application of the interferometer in different domains of atomic physics, astronomy and astrophysics, light scattering, and a final compendium that includes metrology, optical bistability, velocimetry, the infrared, sensors, plasma physics and miscellaneous devices. It is obviously not possible to provide in these chapters a complete and comprehensive account of the role of Fabry–Perot interferometry in all these subjects — each alone would require a volume to itself. The aim has been rather to outline chronologically the more significant developments and to describe and illustrate techniques of particular utility. It may be remarked how often techniques almost standard in one region of a subject, are subsequently ‘reinvented’ in another. It may be hoped that the present work will assist such cross-fertilization. The chapters are supplemented by a number of appendices on material which I judged to be useful but which seemed either to break the flow, or not quite to fit into specific chapters. At

many stages in preparing the text I have been conscious of intriguing and exciting byways of instrumentation and application that called out for examination, analysis and description. All too often such siren calls had to be resisted if the book was to be kept in reasonable bounds.

The literature of Fabry-Perot interferometry is now so vast that there seemed little hope of providing an exhaustive bibliography. However the list of references is reasonably extensive and I have endeavoured to reference material fairly closely in the text. I hope that most topics relating to Fabry-Perot interferometry are either covered in the text or at least are sufficiently referenced as to be quickly found in the original literature. The aim throughout has been to provide useful material for the working spectroscopist, material that is of direct instructional benefit to the tyro and offers perhaps a broader perspective to the experienced practitioner. In many instances specific techniques are described that I have found of particular value; undoubtedly many workers have developed equivalent, possibly superior, methods for such tasks. The problem is that these techniques are rarely described in the literature and are usually passed on by word of mouth (supposing that is that they are not jealously guarded!). The present work seemed a useful medium of at least propagating some of this spectroscopic folklore derived over many years with many colleagues. In part justification of this I would remark that most spectroscopists have experience of optical systems that took two weeks to align the first time, two days the second and eventually could be assembled, routinely, in less than a few hours.

This book is devoted to describing a particular instrument; most spectroscopists are aware of the satisfaction and sense of wholeness that a fine optical instrument provides. Nevertheless, however perfect and however novel, the final justification of any instrument must be in the information it provides and the new knowledge that it leads to — the spectroscopist forgets this at his peril! The early papers of Fabry and Perot are a model in this regard with their lucid exposition of experimental technique and their immediate application to the important scientific problems of the day. This is not of course to decry the skill and importance of instrument designers and builders, or the value of more theoretically inclined studies of for example performance limitation; however the ultimate criterion for any instrument must be its performance in practice on scientific tasks. Fortunately the scientific record of the Fabry-Perot interferometer puts it beyond any such criticism.

Particularly with the vast expansion of optical studies relating to the laser, many workers are now using interferometric techniques on problems quite unimagined until recently. Many of these people are comparative newcomers to optics without the long traditional spectroscopic training of past generations. At least part of the hope and justification of the present work is that it might be of service to them, and that Fabry-Perot interferometers will as a consequence be employed with yet greater understanding and effectiveness.

J M Vaughan
Malvern 1988

Acknowledgments

I am indebted to many colleagues for references and source material and particularly thank Dr B Bates of Queens University Belfast, Dr N Berovic and Professor W F Vinen of Birmingham University, Professor D Burgess of the Blackett Laboratory, Imperial College, Dr C Coleman of the Marconi Company, Dr J Hough of Glasgow University, Dr David A Jackson and Dr J C D Jones of the University of Kent, Dr W H King and Dr E L Lewis of Newcastle University and Dr G Smith of the Astrophysics Department at Oxford. I have received help and advice from many colleagues at RSRE with figures, references and constructive criticism of parts of the manuscript and express my appreciation in particular to Derek Brown for preparing all the computer plotted figures, and also to Robert Brown, Hywel Davies, Michael Harris, Robin Jones, Terry Mason, Alan Miller, Roy Pike, David Willetts and John Wilson. As always any errors or omissions are the sole responsibility of the author. Many illustrative figures taken from the scientific literature are referenced and attributed in the text; the permission of publishers to use these is gratefully acknowledged.

I am greatly indebted to Roy Pike as Series Editor, and also to Jim Revill, Commissioning Editor at Adam Hilger and Jen Halford, Desk Editor, for their advice and assistance. I am also grateful to Mrs Mollie Johnson and Sarah and Judy for listing and checking the many references and for preparing the index. Finally I would like to thank Mrs Mick Roberts who typed the bulk of the manuscript and with great good humour converted pencilled scrawl into legible copy.

1 Historical Background

1.1 Introduction

In the last years of the 19th century two young French physicists at the University of Marseilles, Alfred Perot and Charles Fabry, described a novel form of interference device. This made use of interference phenomena due to waves successively reflected between two thinly silvered plane glass plates set accurately parallel. A simple theoretical analysis of this problem had in fact been given over sixty years earlier by George Airy, an English astronomer. Fabry and Perot swiftly realized that the sharp, multiple-beam fringes from their equipment offered great precision for size measurement, wavelength comparison and a wide range of applications in metrology and spectroscopy. In a series of brilliant papers, mostly in French but often with simultaneous appearance in English in the *Astrophysical Journal*, they developed the underlying principles, described several superbly constructed interference instruments, introduced and evaluated experimental methods and techniques and illustrated the potential applications with thorough measurements. Comparable instruments of the time were the diffraction grating which had been brought to its highest pitch in the work of Rowland, and the two-beam interferometer of Michelson. Whilst of great general utility the absolute accuracy of the wavelength standards introduced by Rowland were in fact coming under increasing criticism. Fabry and Perot added to this and supported the comparative measurements of Michelson in which the 'interference' was more akin to their own device. This view itself brought the justified criticism that all these instruments, including the grating, relied on interference. However for the most part Fabry and Perot defended the advantages, real and potential, of their instrument with great clarity and vigour. Within ten years all the major applications and techniques had been introduced, largely from Marseilles, and remained standard practice for over fifty years. Several instruments were soon constructed in laboratories around the world—notably in Germany, Great Britain and the United States. After some initial difficulties most experimentalists commented favourably on the reliability of their equipment and the ease of interpretation of the measurements. Further development was held back by the Great War, but afterwards an increasing flow of spectroscopic measurements with the now widely named 'Fabry-Perot interferometer' were reported (often in France 'Perot-Fabry').

Some of the later history is outlined briefly in appropriate chapters throughout the book; this introductory chapter describes the early years following Fabry and Perot's first papers. It is not the intention of this chapter to provide a history of spectroscopy during this time. It is rather to focus quite closely on the early papers of Fabry and Perot; these in many ways give a remarkable insight into the development of a technology. Perhaps as importantly, in their blend of theoretical analysis, physical insight, description of equipment, and practical exploration the papers themselves provide a masterly guide to scientific investigation.

1.2 The Airy distribution

In 1831 George Biddell Airy, Plumian Professor of Astronomy in the University of Cambridge brought out a second edition of his '*Mathematical Tracts*'. For this edition he had added a tract on 'The Planetary Theory' and one on 'The Undulatory Theory of Optics'. The book was described as 'Designed for the Use of Students in the University', and in the preface to the new edition Airy wrote: 'The Undulatory Theory of Optics is presented to the reader as having the same claims to his attention as the Theory of Gravitation: namely that it is certainly true, and that by mathematical operations of general elegance it leads to results of great interest'. The rest of the preface is in fact a spirited defence of this position†. In introducing the Tract itself (p249) Airy began with:

ON UNDULATIONS GENERALLY.

PROP. 1. To explain the nature of an Undulation.

1. The characteristic of an undulation is, the continued transmission in one direction of a *relative state* of particles amongst each other, while the motion of each particle separately considered is a reciprocating motion. The disturbance of the particles from their state of rest, and their motion, may be in any direction whatever.

After detailed discussion of wave properties Airy came to:

PROP. 5. To explain the interference of undulations.

9. By *interference* is meant the co-existence of two undulations in which the length of a wave is the same. The

†Airy was a leading proponent of the undulatory or wave theory of optics and subsequently published a lengthy article on Newton's rings in volume II of the *Philosophical Magazine* in 1833. An interesting sidelight on the continuing controversy over wave versus corpuscular theory may be found in this latter volume in the increasingly acrimonious correspondence with a Mr R Potter Jr (pp161, 276 and 451). See also the critical remarks by Sir David Brewster (p360), Airy's reply on p419, and several other papers by eminent authorities of the day.

conception of interference is not in any circumstances easy†, and it is more particularly difficult with regard to Physical Optics, from our ignorance of the physical causes to which the undulation is due.

His plaintive comment that the conception of interference was not easy surely finds echoes amongst students to this day! However he promptly provided the illustration of two waves on the surface of water: 'it may be imagined that . . . the elevation of one wave may exactly fill up the hollow of the other . . .'. In the following pages he developed mathematically ideas of interference, propagation of a wave 'by . . . dividing it into an indefinite number of smaller parts . . .', and refraction and reflection. Following this Airy calculated several interference problems including light reflected from two slightly inclined mirrors, transmitted through a small prism, and with interpolation of a thin piece of glass. These led to the important calculation and formula which bear his name and which stem from the following propositions:

PROP. 15. A series of waves are incident upon two plates of glass separated by a very small interval (fig. 17) part of the light is reflected at the lower surface of the first glass and part at the upper surface of the second glass: and these portions interfere: to find the intensity of the mixture.

and

PROP. 16. In the circumstances of the last proposition, to find the intensity of the light refracted into the second plate.

A direct copy of Airy's figure 17 is shown as figure 1.1. In an elegant derivation Airy calculated the retardation V of successive components of vibration as being $2D \cos \beta$ and showed that the brightness of the reflected light was

$$4a^2e^2 \sin^2 \frac{\pi V}{\lambda} \left(1 - 2e^2 \cos^2 \frac{2\pi V}{\lambda} + e^4 \right)^{-1} \quad (1.1)$$

and of the transmitted light was

$$a^2(1 - e^2)^2 \left((1 - e^2)^2 + 4e^2 \sin^2 \frac{\pi V}{\lambda} \right)^{-1}. \quad (1.2)$$

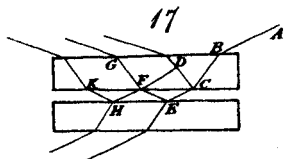


Figure 1.1 From Airy (1831, figure 17). The incident ray is shown entering along AB and experiencing successive reflections and refractions at C , E , F , H and K . The construction line FD is drawn perpendicular to the ray path CD .

In these expressions a and e are the vibration coefficients for the incident light and the air-to-glass reflection respectively and proper allowance was made for phase changes. In the manner that is the continuing glory of physical optics he immediately illustrated these mathematical expressions with examples: (i) '... when a soap bubble has arrived at its thinnest state, just before bursting, the upper part appears perfectly black'; and (ii) 'The intensity is also 0 if $D \cos \beta = \lambda/2, \lambda, 3\lambda/2$ etc, but ... it will be impossible to make the light of all the different colours vanish with the same value of D , and thus no value of D will produce perfect blackness'. In transmission Airy considered two close spaced glass prisms with light passing between them close to the angle of total reflection so that β may be made close to 90° . He noted that in this situation, for small changes of the angle of incidence, 'the expression for the intensity of light will be varied much', and in consequence 'if the sun light be made (by a lens) to fall on it in different directions ... the position and breadth of those bands is different for every colour, the mixture forms a very splendid series of coloured bands, in which the succession of colours differs from that produced by almost every other phenomenon of interferences. ... if the combination of prisms be held to the eye ... the light ... will exhibit the bands in great perfection'.

In the next Proposition (18) Airy applied his formula to Newton's Rings, justifying this on the basis that the two surfaces are so close that 'the defect from parallelism will produce an error amounting only to a very small fraction of λ '. Airy calculated the diameters of the rings for this situation in terms of β , λ and the radii of curvature of the surfaces and noted again the complementary character of reflected and transmitted light. It is particularly interesting however that he did not make any mention of the corresponding possibility of circular fringes, for plane parallel surfaces, arising from the cylindrical symmetry of his basic expressions. One must suppose that he did not recognize this possibility, and indeed their existence in soap films for example would not have been easy to observe.

The first observation of complete rings must be credited to W Haidinger who described them in 1855. In an earlier note in 1849 he discussed fringes seen in reflection in a parallel sided mica plate at considerable obliquity—these were approximately straight bands. In 1855 he described circular rings obtained in reflection with light around normal incidence. These circular fringes were viewed through a glass plate inclined at 45° which served as a beam splitter, passing the incident light to the mica plate and transmitting the reflected light to the eye. The much weaker circular fringes were also seen in the transmitted light.

Many years later there seems to have been some controversy as to whether Haidinger actually understood the character of the fringes he was observing. Lord Rayleigh had no doubts; he duplicated, described and extended Haidinger's original arrangement and in 1906(b) wrote: 'The importance which these rings have acquired in recent years owing to the researches of (Lummer), Michelson, and of Fabry and Perot, lends interest to the circumstances of their discovery. It seems to be usually supposed that Haidinger merely observed the rings without a full appreciation of the mode of formation. ... A reference to

the original papers will, I think, show that Haidinger, in spite of one or two slips, understood the character of the rings very well, and especially the distinction between them and the rings usually named after Newton and dependent upon a variable thickness in a thin plate'.

What cannot be disputed however is the first truly spectroscopic observation and application of Airy's formula with light of the two sodium D lines. This must be credited to M R Boulouch who in 1893 considered the fringe systems of rings formed simultaneously for two wavelengths λ and λ' in both transmission and reflection using Airy's expressions. He wrote†: 'Il est possible de voir simultanément les deux systèmes de franges si l'on se place dans des conditions convenables fournies par l'examen des formules d'Airy' [1]. His calculations (carried out for the reflected light intensity $I + I'$) showed quite clearly that, for a phase differential between the systems of $\pi, 3\pi, \dots, (2n + 1)\pi$, and for the reflection coefficient h close to unity, '... il est aisé de voir que $I + I'$ présentera deux maxima identiques séparés par des minima peu différents l'un de l'autre mais jamais nuls; le nombre des franges sera doublé.

On obtiendrait des résultats identiques pour la lumière transmise puisque les anneaux transmis sont complémentaires des anneaux réfléchis' [2].

He noted that h could be made large in a glass plate at grazing incidence and most importantly near normal incidence: '... il suffira de produire des anneaux entre deux lames transparentes recouvertes d'une couche d'argent assez mince, pour que l'ensemble soit un peu transparent pour la lumière du sodium; en éloignant deux surfaces, les anneaux dédoublés apparaissent, pour une épaisseur convenable de la lame, singulièrement nets' [3]. However Boulouch gave rather few experimental details—what plates he had used, how they were held apart and how they were adjusted—and does not seem to have taken the technique further, other than noting that it could be extended to examine flames of lithium and thallium. Perhaps in consequence this remarkable paper has been somewhat neglected over the years.

A fascinating connection with this historic period was recently given by Professor Alfred Kastler the eminent French physicist and Nobel Laureate. In unpublished notes‡ for a book review, written shortly before his death at the age of 81 in January 1984, Professor Kastler recorded that when he (Boulouch) wrote his paper: 'Boulouch was at that time a high school teacher at the Lycée de Bordeaux where I myself was teaching in the years 1929–1931 and where I met the old Boulouch, retired at this time. Charles Fabry, before being nominated to Marseille University, was during the year 1889–90 teaching at the Lycée de Bordeaux where he had been Boulouch's colleague. But he never mentioned the anteriority of Boulouch's investigation. To my knowledge the

†In this introductory chapter we will make frequent use of the original words of the authors in their own language. An English translation of these numbered passages will be found at the end of the chapter.

‡I am indebted to Professor E R Pike for this valuable historical information.

only one who cites Boulouch's work is S Tolansky, but giving a wrong reference (1906 instead of 1893)'.

To be strictly accurate Boulouch's work was referred to, once, by Fabry and Perot on page 469 of their first paper of 1897. They wrote: 'Les *dédouplements* des franges fournies par la lumière du sodium ont été déjà décrits par M. Boulouch (1); ils remplacent les disparitions découvertes par Fizeau dans le cas des lames minces ordinaires' [4].

The only other reference in the literature of the succeeding three decades appears to be that of Feussner and Janicki in 1927.

1.3 The early work of Fabry and Perot

In the forty years following Airy and Haidinger many significant advances were made in optics with contributions from most of the notable physicists of the 19th century. (For a brief review see for example the historical introduction to *Principles of Optics* by Born and Wolf (1975)). The electromagnetic wave theory of Maxwell, and verification in 1888 by Hertz that such electromagnetic waves were indeed light waves, put the subject on a satisfactory basis. In this regard of course it answered Airy's comment on '... ignorance of the physical causes to which the undulation is due'. In December 1897 Fabry and Perot published in *Annales de Chimie et de Physique* their paper 'Sur les franges des lames minces argentées et leur application à la mesures de petites épaisseurs d'air' [5]. This work arose from their interests in electricity and the necessity therein of precise measurements of small distances. They noted prophetically that these '... nous ont conduits à l'étude de phénomènes d'Optique intéressants en eux-mêmes, et à l'invention de méthodes de mesure des petites épaisseurs qui pourront sans doute rendre des services dans d'autres cas' [6]. Fabry and Perot firstly pointed to the sharpness and contrast ratio of fringes arising in a thin film of air between glass plates. They showed that the trough to peak ratio ρ equalled $[(1-f)/(1+f)]^2$ and illustrated this with silvered plates of reflection coefficient $f = 0.74$ for which ρ is equal to 0.02. After referring to Airy they provided a derivation of the Airy expression in transmission of the form given (after correcting a trifling error in their equation (2)) by

$$I = \frac{\theta^2}{(1-f)^2} \frac{1}{\{1 + [4f/(1-f)^2] \sin^2 \pi \Delta/\lambda\}} \quad (1.3)$$

where now θ and f are transmission and reflection coefficients and Δ is the optical retardation of successive components. They commented on the sharpness of the fringes and noted for the term $4f/(1-f)^2$ in the denominator: 'Ce facteur peut être considéré comme définissant la finesse des franges.' [7]. The authors tabulated this factor and also ρ for different values of f between 0.1 and 1. While noting the practical limit with f approaching 0.90, the value for the solid metal, they perceptibly remarked that but for absorption in the silver films the intensity of the maxima would be equal to that of the incident light. They also discussed the types of fringes including fringes localized in the plate '...

chaque frange dessine un lien d'égale épaisseur de la lame mince. Leur forme depend donc absolument de la forme des surfaces ...' [8]. However the following description encapsulates the particular characteristics of their discovery and is worth quoting in full [9] (reproduced by permission of Masson SA):

On

peut, comme les franges des lames minces ordinaires, les observer dans le cas où les surfaces sont exactement planes et parallèles (¹), en visant à travers le système avec une lunette ou l'œil accommodé pour une distance infinie. Les franges, qui ont toujours l'aspect de lignes très fines, se présentent alors sous forme d'anneaux, *localisés à l'infini*, dont le centre est dans la direction normale aux surfaces qui limitent la lame mince. Les diamètres de ces anneaux diminuent à mesure qu'augmente l'épaisseur de la lame qui les produit.

Having thus characterized the salient features of Fabry–Perot fringes: plane parallel surfaces, circular fringes, localized at infinity, reducing in size for larger spacing, the authors in the remainder of the paper emphasized the measurement of small thickness with, for the most part, fringes localized at the plates. They showed an optical arrangement for projecting the fringes onto a screen and described how coincidence techniques might be used to determine the order numbers of fringes of different spectral lines derived from metal arc discharges. They emphasized that the narrowness of the fringes allowed of course a very precise comparison and assignment. In an imaginative step they then considered the fringes of superposition in a beam of white light traversing *two* thin air plates. This provided them with a straightforward technique for measuring small thicknesses; rapid and precise comparison could be made with a standard calibrated plate with a narrow wedge angle. This device they called a 'lame—étalon' (lame—film or layer; étalon—standard of weight or measure); they described a model made up of glass plates 20 cm long, 3 cm wide and 1 cm thick, separated by thin strips of mica of slightly different thickness to form the wedge, and bonded with mastic. In another ingenious device they described a mobile plate constrained between two fixed plates, all silvered. By observation of the superposition fringes between the two air spacings, movements as small as $\lambda/200$ of the central plate could be observed; in addition the central plate could itself be readjusted to a set position without the necessity of following its displacement and counting fringes.

In §IV of their paper Fabry and Perot measured the small change of phase on reflection at the silver films and noted this as a very small correction factor for their measurements. Finally they provided a detailed explanation of the interference rings due to scattering in the silver surfaces when observing a bright source. Clearly a remarkable profusion of ideas was revealed in this first joint paper.

The birth of the Fabry–Perot interferometer has been variously ascribed to

this article and the year 1897, and also to 1899 when, in several papers, they described further studies of interference. Indeed one of these latter papers describes an equipment that is clearly an interferometer (see Appendix 1). Nevertheless, the broad comprehension of the properties of the multiple-beam fringes that carry their names is so obvious in this first paper that it would seem appropriate to consider it as the real beginning.

In a related development in the following year it is worth noting that Edser and Butler, working in London, described their well known ‘plate’ for the rapid and simple calibration of a prism spectroscope. This relied on fringes formed with white light illumination in a narrow air film with successive fringes given (μ refractive index, d film thickness) by the relationship

$$2\mu d = n\lambda_0 = (n + 1)\lambda_1 = (n + 2)\lambda_2 = \dots = (n + m)\lambda_m. \quad (1.4)$$

Edser and Butler referred directly to Fabry and Perot’s paper and the advantage of silvered surfaces and the consequent sharp fringes. Their own device was quite simply constructed: ‘... if a little soft wax be placed around the edges of the plates a perfect adjustment can be obtained by simply pressing the plates together with the fingers’. After bringing successive images of a small source into coincidence final adjustment was made by broadening the interference bands from a sodium flame. In a photographic spectrum shown as an example, over 400 white light fringes were recorded between the sodium line at 5890 Å and the calcium line at 3968.6 Å.

1.4 The papers of 1899

Early in 1899 Fabry and Perot published four papers on their work on interference. Two of these, in *Annales de Chimie et de Physique*, were lengthy and detailed accounts: ‘Théorie et applications d’une nouvelle méthode de spectroscopie interférentielle’ (January), and ‘Méthodes interférentielles pour la mesure des grandes épaisseurs et la comparaison des longueurs d’onde’ (March) [10]. The third paper appeared in the January edition of *Bulletin Astronomique* and was in effect a summary of the two other papers reduced to somewhat less than a half. This paper also appeared, in direct translation into English, in the *Astrophysical Journal* with the title ‘On the application of interference phenomena to the solution of various problems of spectroscopy and metrology’. Consideration of the section headings of this fourth paper gives an indication of the thoroughness of their studies: I. Fringes from silvered plates; II. Phenomena produced when the incident light is composed of two monochromatic radiations; III. Application to spectroscopy; IV. Determination of the order of a fringe; V. Comparison of wavelength; VI. Measurement of lengths. However, only by detailed reading, particularly of the two longer papers, can a full appreciation be made of the depth, quality, and insight displayed in this large body of work.

Of particular interest is the instrument they describe. Having established that the fringes should be observed with a telescope focused for parallel rays, they

noted that their diameter obeyed the same law as for Newton's rings with the difference that the optical retardation attained a maximum at the centre. They specified the technical requirements for the instrument they had constructed as follows: 'These surfaces which face one another, must be made exactly parallel; it is convenient to be able to change their distance apart without destroying this parallelism, so that without readjustment it may be possible to observe the rings corresponding to various values of the difference of path'. Would that such a specification could be routinely achieved in this day!

Appendix I shows in facsimile the detailed description given by Fabry and Perot in their January paper in *Annales de Chimie et de Physique*, with two cross sectional drawings. (A perspective view of the highly engineered version with the same basic layout, built for them by 1901, is shown in figure 1.4.) The shortened description appearing in English in the *Astrophysical Journal* of 1899(a) went as follows (reprinted courtesy of the Astrophysical Journal, published by the University of Chicago Press; © (1899) The American Astronomical Society):

The apparatus employed, which we call an *interference spectro-scope*, essentially consists of two plane plates of silvered glass placed vertically. One of these plates, L , carried by an old theodolite, can be given a wide range of displacement in azimuth, and small parallel displacements by means of a strong iron stirrup which can be bent by distending a rubber bag filled with water placed inside the stirrup; this bag is connected, like the two others referred to below, by a rubber tube to a vessel filled with water, the level of which can be varied; it is thus possible to produce a displacement of a few microns by a motion as slow as may be desired, and without lost motion.

The other plate, L' , can be given small adjustments in azimuth and large parallel displacements. It is carried normally at the end of a horizontal strip of steel, 5mm in diameter and 10cm long, rigidly fixed at the other end, against which two bags filled with water press in two directions at right angles to one another; it is thus possible to obtain through flexure of the steel strip, very small angular displacements which are produced without in any wise disturbing the apparatus. The steel strip is supported by a carriage having the form of a triangular prism with horizontal edges; two strips of St. Gobain glass are cemented to the two lower faces, which meet at a right angle; these rest on ways also made of glass strips cemented to a wood base. The carriage may thus be given a parallel displacement, in which it will be perfectly guided if not subjected to any lateral pressure. To effect this the carriage can be pushed, in either direction, by two points attached to two other auxiliary carriages; the principal carriage has a little play between the two points, and consequently can be pushed by only one of them. The two auxiliary

carriages can be moved together by means of a screw connected with them, which passes through a nut that can be turned by hand when a rapid displacement is desired, and by means of a tangent screw when a slower motion is needed. A single turn of this screw corresponds to a displacement of 3μ or 4μ . The whole apparatus, carried on a thick plank, is suspended in the air by rubber rings to protect it from vibration, which would render the fringes invisible.

The adjustment of the parallelism of the plates is effected by moving the plate L for approximate adjustment, and L' , through flexure of its support, for the final adjustment. Large parallel displacements are produced by moving the nut on the thread, and small ones by bending the stirrup which carries the plate L . Accidental displacements, caused by imperfections in the guides along a distance of several centimeters, are so small that the rings do not disappear, and they can be brought back to their normal appearance by bending the support of L' .

A divided scale, attached to the plate-carriage, and read by a fixed microscope, gives a means of measuring approximately the displacement of the carriage, and consequently the distance between the silvered surfaces.

To the modern instrumentalist the startling aspects of this the first 'Fabry-Perot interferometer' must be the ability to scan the fringes over several orders, the coarse and fine control of plate separation, the provision for rapid and variable change of plate separation while maintaining near parallelism, and the protection against vibration.

Underlying much of the use of their interferometer was the phenomenon of 'coincidences' which they described in considerable detail. Consider the particular fringes shown in figure 1.2 where A and A_1 are consecutive red (λ) fringes of order K and $K + 1$ and between them occur yellow (λ') fringes B and B_1 of order K' and $K' + 1$. The order numbers K and K' are obviously integers and we suppose $K' - K = m$. With the fringes as shown, for some *one* position between B and B_1 the *difference* of the (non-integral) order numbers for each set of fringes must be m and be given by

$$(\Delta_0/\lambda') - (\Delta_0/\lambda) = m \quad \text{and} \quad \Delta_0 = m(\lambda\lambda')/(\lambda - \lambda') \quad (1.5)$$

where Δ is the optical path difference for consecutive interfering beams. The phenomenon is periodic, repeating itself for values of Δ which are multiples of the length Π given by $(\lambda\lambda')/(\lambda - \lambda')$ which Fabry and Perot called the 'period'. If A and B overlap then the coincidence may be said to be *exact* with the K th red fringe matching the $(K + m)$ th yellow fringe. More generally the appearance will be as shown in figure 1.2 with the coincidence at order $K + \theta$ for the red fringe system with $0 < \theta < 1$.

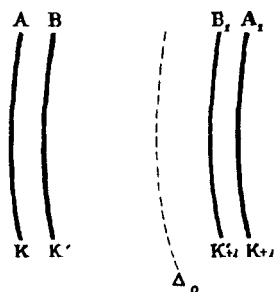


Figure 1.2 Interference fringes for a pair of radiations A and B, showing the order numbers K, K' etc. At some intermediate position of path difference Δ the difference of order number will be the integer m equal to $(K' - K)$, thus giving a coincidence. (From Perot and Fabry 1899, figure 1.) Reproduced by permission of Masson S A.

Fabry and Perot applied this concept of coincidences very extensively in their first group of papers. For the spectroscopic study of a close group of lines they noted that experience had shown that 'the separation of the two systems of rings is always clearly visible when the distance between rings of different kinds is $\frac{1}{5}$ the distance between consecutive rings of the same kind'. Thus for two radiations of nearly equal wavelengths λ and $\lambda + \epsilon$ they noted that it was sufficient to reach an order number p corresponding to $\frac{1}{5}$ the period i.e. $p = \lambda/5\epsilon$. This would be achieved with silvered surfaces of separation e equal to $p\lambda/2 = (\lambda/10)(\lambda/\epsilon)$. From this they remarked that for e equal to 5 mm it was possible to resolve lines 1/1000 of the distance between the sodium D lines—a power '... already comparable with that of the best spectroscopes having prisms or gratings'. For approximately monochromatic light, as the plate separation increased, they commented that 'Each fringe constitutes a veritable spectrum of the luminous source.'

Fabry and Perot extended their studies to lines in metallic vapours—mercury, cadmium and thallium, and noted that their results were not identical with those deduced by Michelson and from fringe visibility measurement. However they rushed to add: 'In fact, our results would lead to visibility curves identical with those of Professor Michelson; far from contradicting the experimental results of this investigator, our researches confirm them completely'. They also noted how their method was readily adapted to studying the change of wavelength of a given line: '... attention being directed to the change in appearance of the rings produced by the two sources successively'.

In addressing the problem of determining the order of a fringe Fabry and Perot made ingenious use of their method of coincidences. By comparing carefully chosen radiations combined in pairs they reduced the observation to obtaining three integral numbers. For thicknesses up to 4 or 5 cm, sets of less than ten coincidences were required to be counted to provide a rigorous determination of fringe number that could be as high as 200 000. They

emphasized that for this method the ratios of the wavelengths employed needed to be known very precisely; a precision of 10^{-6} was not sufficient: ‘For the lines of cadmium the ratios of wavelengths are known with a precision which leaves nothing to be desired, thanks to the beautiful investigations of Professor Michelson’. This was not true of the mercury lines which led them immediately to make a comparison with the cadmium wavelengths. The technique relied on successive approximations, starting with coincidences of small order number for fringes with small plate separation (typically 1 mm), and building up to larger separation. In this way successive corrections could be made. For 3 mm separation of the silvered surfaces a precision of 1 in 200 000 was obtained which they noted would be difficult to surpass with a grating instrument, particularly for widely separated lines. These initial measurements were extended to 32 mm separation; the quoted results for air at 15 °C and 760 mm pressure were:

Yellow lines of mercury $\lambda_1 = 0.579\,065\,93\,\mu\text{m}$

$\lambda_2 = 0.576\,959\,84\,\mu\text{m}$

Green line of mercury $\lambda = 0.546\,074\,24\,\mu\text{m}$

with a probable error of 5 units in the last place, or one part in 10^{-7} . In the greater detail of the *Annales de Chimie et de Physique* papers the fine structure in these lines and in blue and green lines in thallium and cadmium is described. The wavelengths quoted are those of the principal components. Figure 1.3 from their March paper shows the manner of recombining two sources T and T' with the beam splitter A. Fringes formed by the interferometer I are observed by the telescope. It is interesting to note the manner of illumination of the interferometer—Fabry and Perot refer many times to the ‘convergent light’ required to illuminate the fringes localized at infinity. This presumably arose from the necessity to observe a large number of rings with relatively small sources.

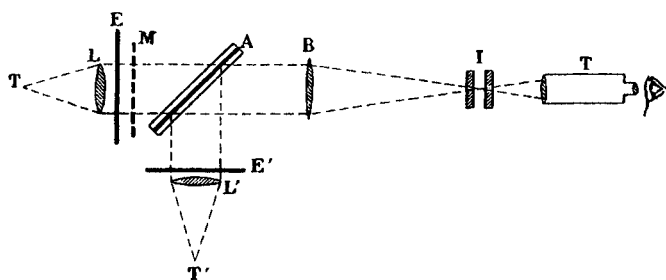


Figure 1.3 The experimental arrangement for combining beams from the sources T and T'. Note the ‘convergent light’ on the interferometer I. (From Perot and Fabry 1899, figure 3.) Reproduced by permission of Masson S A.

In the measurement of length Fabry and Perot maintained that with superposition fringes it was possible to fix a reference position within a few thousandths of a micrometre. For measurement of large thicknesses they employed convergent light with a telescope focused for parallel rays and plates

of uniform spacing. They described very precise techniques using white light fringes for measuring small differences of thickness and, given an adjustable plate, of creating an exact equality so that a given thickness could be copied. This technique could be extended to creating lengths a factor of 2, 3 and up to 4 times larger or smaller. Since they were able to determine the number of monochromatic fringes within a 4 or 5 cm thickness this was immediately extendable to 20 cm in one step; they suggested that ‘... it seems possible in good conditions to observe superposition fringes with thickness of at least 1 m’.

In §VII of their January paper in *Annales de Chimie et de Physique* they considered the fringes due to a single glass plate with silvered parallel surfaces and derived an expression taking account of dispersion in the glass. M Jobin made a demonstration model for them—a plate of thickness 3 mm, silvered on both sides and mounted in a protective case, which showed clearly fringe doubling with light from thallium. They proudly commented that ‘le pouvoir dispersif de ce petit appareil est par suite de même ordre que celui des meilleurs spectroscopes à réseaux ou à prismes’ [11]. Following this they remarked that for an air-spaced interferometer one should ensure that the plates themselves should have an appreciable wedge (they used 1’) so that fringes due to reflections within the plates would not confuse the principal system produced ‘... par la lame d’air’.

In further technical points they noted the appearance of reflection fringes and also the technique of examining different small regions of a pair of plates to look for changes of ring diameter and hence variations in spacing: ‘... on peut ainsi apprécier des variations d’épaisseur de moins de $\frac{1}{20}$ de μ ’ [12]. Finally they recorded the application to the study of absorption lines which they observed in the sun’s spectrum.

Considering this body of work more than eighty years later one must be impressed by its truly comprehensive character. So many principles of instrumentation, investigation and application are dealt with that to this day are of active concern to experimentalists.

Perhaps the best illustration of this is the fact that it is hard to conceive what other aspects they could have touched on at this stage. Possibly photographic recording of fringes could have been considered and one suspects that with their emphasis on ‘convergent light’ they had not quite fully appreciated the different techniques of introducing light into their interferometer and its large potential light grasp. But these are trifling criticisms. For what other instrument, admittedly introduced against a background of over 50 years of tremendous vigour in the study of interference, can one point to such a thorough evaluation of technique and potential application in such a small group of papers? Indeed their published work that year extended to two actual applications—measurements undertaken with Macé de Lépinay and reported in *Comptes Rendus*. Applying interference methods they determined firstly the size in wavelengths of a cube of quartz and secondly the mass of a decimetre cube of water. One can perhaps best finish this account for 1899 with their concluding paragraph from the *Astrophysical Journal* (reprinted courtesy of *The Astrophysical Journal*, published by the University of Chicago Press; © (1899) The American Astronomical Society):

Such are, in brief, the principal applications that we have made of interference phenomena given by silvered plates. The greater part of the problems that we have attacked have already received other solutions; our methods are notable for the simplicity of the apparatus, the work of the constructor being reduced to the figuring of two plane surfaces. This is evidently a good means of avoiding systematic errors, and of utilizing as completely as possible the remarkable power of interference methods. Particularly for the measurement of lengths and all allied problems, these methods can render effective service with the ordinary resources of a laboratory. The difficulties resulting from the extreme minuteness of the wave-length are largely offset by the precision of the measures and the certainty offered by the standard employed.

1.5 The work of 1900–1902

After the remarkable advances of 1899 Fabry and Perot appear to have devoted much of the following year to investigating various types of light source. In their paper in *Journal de Physique* (1900a) they described excitation in flames and different forms of electrical discharge while cautioning that ‘... il n’est pas possible de parler du spectre d’un corps sans spécifier les conditions de production’ [13]. While discussing the electric arc they offered the comment that ‘Les spectroscopistes de l’école américaine font un fréquent usage de cette source, en France elle est beaucoup moins répondeue et mériterait d’être plus employée ...’ [14]. They went on to describe different forms of arc discharge together with absorption filters for isolating various lines; finally they gave a catalogue of their latest wavelength measurements in mercury, zinc, silver and copper.

In the following year (in a paper in *Annales de Chimie et de Physique*, also in translation in the *Astrophysical Journal*) they described their new form of interferometer—ordered from M Jobin ‘... which has been constructed in a most perfect manner ...’. This equipment, as illustrated in a later publication, is shown in figure 1.4 and is similar in basic layout to the prototype described in Appendix 1. The division of adjustments and controls between the two interferometer plates showed a sensible approach to the problems. Referring to figure 1.4, the right-hand plate had coarse (quick) adjustments of orientation and very small parallel displacements. This latter movement was obtained as before by adjusting the pressure of water in a rubber bag between two steel bars 16 cm long, 2 cm wide and 4 mm thick. The bars were connected together at their ends, while at their mid-points one bar was connected to the carriage and the other was fixed to the plate holder. A change of 1 cm in pressure height produced a parallel displacement of the plate of $0.15\ \mu\text{m}$; up to $20\ \mu\text{m}$ movement was available. The other plate holder could be given fine adjustments

of orientation and large parallel displacements, either rapid or slow. The former were achieved with two rubber bags pressing at right angles to each other; a pressure height change of 1 cm produced an angular displacement of 0.25 s. They noted that 'To obtain perfect parallelism it is frequently necessary to adjust the height of the funnels within a millimeter'. The large parallel displacement was obtained by motion of a central carriage (made in bronze) on two well separated bearing surfaces '... worked with great accuracy'. This principal carriage received '... only such impulses as are exactly longitudinal' from two smaller carriages at either end which were themselves adjusted to the required position by means of a Cardan suspension. As before, to prevent vibration, the apparatus was carried on a small table hanging by four rubber cords.

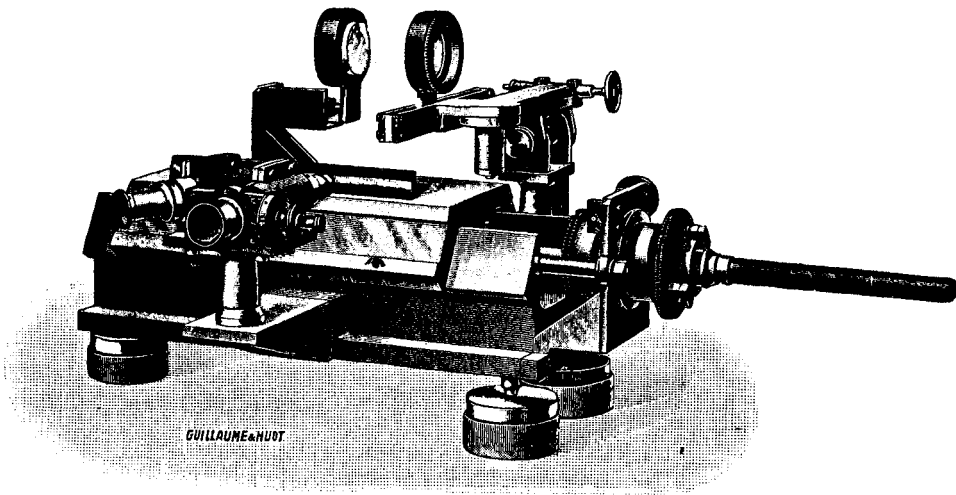


Fig. 20.

Figure 1.4 The highly engineered interferometer built by M Jobin for Fabry and Perot and described and illustrated in 1901. This is the same basic layout as the prototype instrument described in detail in 1899 (see Appendix I). (From Fabry 1923, figure 20.) Reproduced by permission of *Editions de Physique*.

This paper is particularly notable for showing the first photograph of interference rings—for the green mercury line at $\lambda = 5460.7424$ given by a mercury arc in vacuum (see figure 1.5). The authors pointed to the satellites accompanying the main bright ring: a faint interior ring, a second much fainter interior ring and finally an exterior ring which was double. An editor's footnote in the *Astrophysical Journal* regretted that in the published photograph: 'Unfortunately the delicate details of the original photograph are lost in the reproduction'. The authors emphasized that, while the existence of such complex structure had been announced previously by Michelson, this was the first

occasion that the lines had been seen and their positions established. As a further point in favour of their interferometer they concluded: ‘The resolving power . . . increases with the distance between the silvered plates, and can thus be increased as long as the rings remain visible; the finer the lines examined the further can the resolution be carried; in other words, the power of the apparatus is in every case sufficient to show the finest details that it is possible to distinguish’.



Figure 1.5 A photograph of interference rings of the green mercury line, shown by Fabry and Perot in 1901 in the *Astrophysical Journal*. The full structure of four satellite rings was lost in reproduction. This appears to be the first published record of Fabry and Perot fringes. Reprinted courtesy of *The Astrophysical Journal*, published by the University of Chicago Press; © (1901) The American Astronomical Society.

In 1902 Fabry and Perot brought out a lengthy paper on ‘Measures of absolute wavelength in the solar spectrum and in the spectrum of iron’. This appeared in the *Annales de Chimie et de Physique* and, translated into English, in the *Astrophysical Journal*. In the first section techniques of wavelength measurement were discussed in some detail. This discussion contrasted the method of *interpolation* between a series of standards when using a grating with that of *direct comparison* with a single wavelength standard using ‘interference methods’. The authors emphasized that there was no difference in principle but pointed to the very different magnitudes of the respective orders of interference—for gratings just a few units but for ‘interference methods’ up to several hundreds of thousands. In this regard the authors related their own techniques, with multiple-beam interference from silvered plates, to an earlier attempt by Macé de Lépinay to make absolute measurements with interference methods, and also to the two-beam interference techniques subsequently employed by Michelson and Morley and by Michelson and Benoit. Thus drawing

from Michelson's work on the choice of a metrological standard they wrote: '... the best metrological standard is also the best standard from the spectroscopic point of view. We shall therefore choose as standard the red line of cadmium ...'. In fact most of their wavelength determinations were made '... by comparison with the green line, which is brighter and more conveniently situated in the spectrum'.

For most of their experimental work they employed fixed étalons or 'standards of thickness' of which a new and more compact form is shown reproduced in figure 1.6 (again a later illustration). Earlier versions had been described in the 1899 and 1901 papers. Of their latest instrument they wrote '... a new and more compact form is shown in Fig. 1; the silvered plates are held against pieces of polished steel, with rounded ends, by springs whose pressure can be regulated by means of butting screws V. By varying this pressure, the deformation of the steel and the glass at the point of contact is very slightly changed, which permits the parallelism of the silvered surfaces to be adjusted, after having been previously made almost perfect by systematically grinding the pieces of steel. The adjustment of these standards is perfectly stable, and their thickness remains almost absolutely constant after successive dismountings and remountings. We have employed three standards, of 1.5, 5, and 10 mm thickness, respectively.'. Almost identical designs can of course be found in teaching laboratories and non-specialist research to this day.

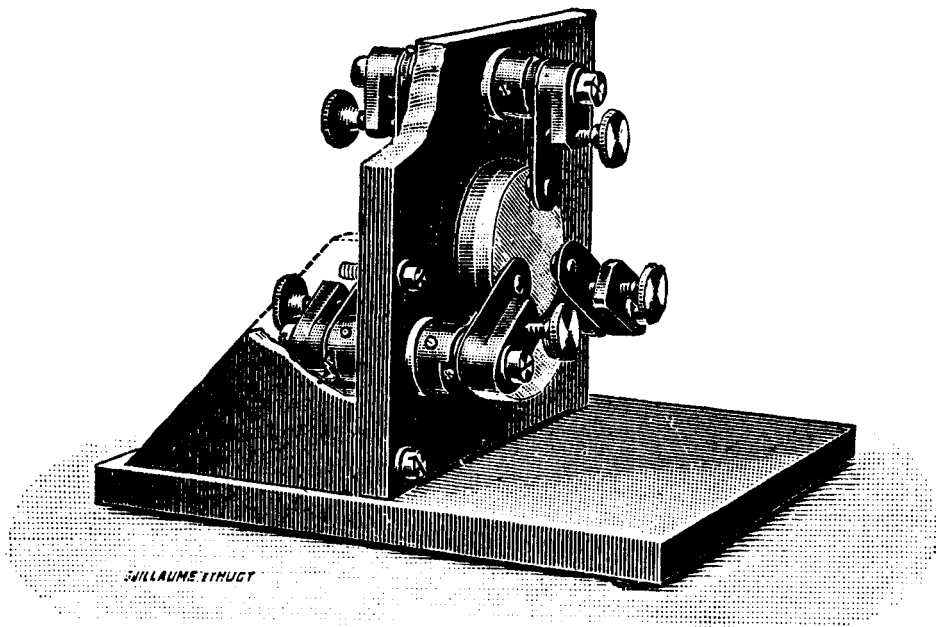


Figure 1.6 A fixed étalon or standard of thickness developed by Fabry and Perot for measurement on arc and solar lines (from the *Astrophysical Journal* 1902, figure 1). Reprinted courtesy of *The Astrophysical Journal*, published by the University of Chicago Press; © (1902) The American Astronomical Society.

For rapid comparison of wavelength Fabry and Perot had by this stage found the method of coincidences rather cumbersome and indeed they listed several disadvantages. They now preferred their 'method of diameters' with a fixed separation of the plates and successive rather than simultaneous illumination by the two light sources. The new method consisted in obtaining for each wavelength the order of interference at the centre of the fringe pattern. Thus for the standard (cadmium) of wavelength λ , if the order of interference of the first fringe from the centre is P (a whole number) and its angular diameter is x , the order of interference at the centre is given by $p = P + \varepsilon$, where $\varepsilon = Px^2/8$. From similar expressions for the radiation to be measured, of wavelength λ' , they derived

$$\lambda(P + \varepsilon) = \lambda'(P' + \varepsilon') \quad (1.6)$$

and

$$\lambda' = \lambda(P/P')[1 + (x^2/8) - (x'^2/8)]. \quad (1.7)$$

At first sight the expressions do not provide a unique value for λ' even when λ , P , x and x' are known. Fabry and Perot pointed out that λ' is usually known with sufficient precision that the actual whole-number value of P' can be selected without ambiguity, and hence an accurate value of λ' obtained. The value of P was obtained for their 'standards of thickness' or étalons from auxiliary measurements with the variable-spacing interferometer. They noted that the 1 cm standard expanded $0.1 \mu\text{m}$ per degree or four fringes for 10° ; however any uncertainty from day to day, arising from change of temperature, was removed by checking coincidences of the red and green cadmium lines (which enumerated the green fringes in fives). In an interesting footnote they commented that the use of steel spacers was advantageous compared with Invar since small temperature changes gave the opportunity of independent measures of sets of x and x' . The angular diameters (x and x') were measured with a telescope provided with a micrometer eyepiece. They noted that for ε , $\varepsilon' \leq 0.3$ small but not insignificant errors of measurements could be introduced since the geometrical middle of a ring and the peak of a ring did not quite coincide. The small corrections due to change of phase on reflection at the silver films were evaluated by inserting the plates in their variable interferometer and checking the small shifts of coincidences with known wavelengths. For the thickest silver surface film of 61 nm (described at the time as $\mu\mu$), they found across the visible band a change of effective thickness e of the standard given by

$$e_{644} - e_{436} = -8.7 \text{ nm}$$

where e_{644} and e_{436} refer to the cadmium red and mercury violet lines respectively. With intermediate measurements at cadmium and mercury green a complete correction curve was obtained.

For their measurements of iron lines from an electric arc in air, secondary dispersion was supplied by a plane grating of Rowland with a single lens as collimator and projector. They wrote: 'To identify the lines we have employed the admirable photographic chart of Kayser and Runge', published in 1888. Of

the 14 lines examined, all were measured with the 5 mm étalon and a few were sufficiently narrow to permit use of the 10 mm standard. The following values are selected from the example they gave:

Wavelength according to Kayser and Runge	561.581
Wavelength according to Rowland	561.5877
λ' adopted in computing P'	561.568
λ' measured by Fabry and Perot (1902)	561.5657.

The maximum relative error was claimed to be about 2 units in the final figure which was also the phase correction with the 5 mm standard. They claimed that there was never the slightest doubt about the whole number P' .

In the solar spectrum, measurements were made on the dark fringes of absorption lines. To achieve sufficiently large dispersion they employed a concave Rowland grating of the largest size (14.5 cm by 5 cm ruled surface, 6.50 m radius of curvature, 568 lines to the millimetre) in second order. A neat optical arrangement was devised in which the slit before the étalon, and the beam path to it from the grating were fixed. The fore-slit to the grating and the illuminating mirror (which received the radiation from a polar heliostat) were set on a movable and easily adjusted carriage. In order to make most effective use of the light passing through the narrow slit, and to illuminate two or three rings adequately, they adopted a complex system of introducing light into the étalon. This incorporated four lenses, three of them cylindrical. Measurements were made with the 2.5 mm standard étalon. They noted that the Doppler shift of wavelength, on passing from centre to limb at the solar equator, amounted to 1 part in 150 000—seven times the measurement precision, and the rings showed an easily visible contraction or expansion in passing from one limb to the other. Measurements were made on 33 lines of the solar spectrum between 460 and 650 nm. The table of results gave not only their own wavelengths λ , but also those of Rowland λ_R , together with the ratio λ_R/λ . This ratio showed systematic variations between 1.000 030 and 1.000 037, the graph of which they maintained provided a correction curve for Rowland's measurements. On any scale of values this paper and body of work must be accounted as truly remarkable. The straightforward description of apparatus and the method of analysis, the obvious attention to detail in the measurements themselves and the requisite correction factors, the imaginative step of making the first solar and absorption line measurements, must, to this day, set for spectroscopists a high bench mark of excellence.

1.6 Progress after 1902

By 1902 the main principles of using interferences between two silvered plates had been established by Fabry and Perot. The technique was being taken up at other laboratories and the potential precision and value of the methods were

becoming appreciated. Alfred Perot had in fact left Marseilles in 1901 to direct the new Laboratoire d'Essais in Paris (see the short biographies on pp vi and vii). Perhaps the one remaining gap in the technology was closed by Fabry in 1905 when he displayed in *Comptes Rendus* the photograph reproduced in figure 1.7. This showed for the first time the simultaneously recorded interference spectra of many lines†. To produce this he wrote: ‘Je place pour cela, dans le plan focal ou ces anneaux viennent se peindre, la fente d’un spectroscopie sans astigmatisme. Cette fente étant laissée un peu large on verra, dans l’oculaire du spectroscopie, chaque raie sous forme d’un rectangle, dans lequel se dessinent les anneaux correspondents. . . .

La figure ci-dessous, qui représente la partie bleue du spectre du titane dans l’arc électrique, donne une idée des épreuves obtenues par cette méthode. L’observation est faite avec une fente large.

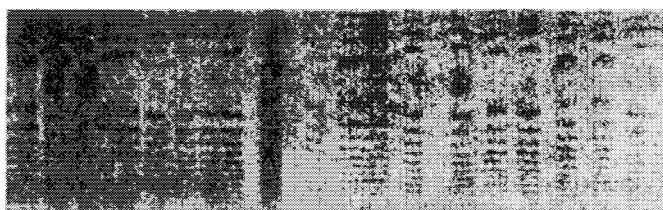


Figure 1.7 The first published photograph of interference spectra taken in the blue region of the titanium spectrum in an electric arc (from Fabry 1905). Reproduced by permission of *Comptes Rendus de l’Académie des Sciences de Paris*.

Dans tous les cas on a une vue ensemble du spectre, et l’on ne peut commettre aucune erreur sur la radiation que l’on examine. Tous les anneaux sont, à la fois, concentrés sur le petit rectangle correspondant, et une seule pose photographique permet de les obtenir tous à la fois. Enfin la lumière est beaucoup mieux utilisée que dans l’ancien procédé’ [15].

For this work Fabry employed a heavy-flint prism spectroscope; he noted also that plane gratings could similarly be employed without difficulty to provide the secondary dispersion. Thus by 1905 the full technique of photographic recording was established which of course remained dominant until photoelectric methods were introduced, also in France, in the late 1940s.

Prior to this final seal on their work Perot and Fabry in 1904 had jointly published a short but magisterial note on the recently described interference plate of Lummer and Gehrke (1903). Using this the latter authors had claimed

†In fact photographs taken simultaneously of a number of lines (mercury yellow and green) were shown by Barnes in 1904—plates 2–5 in figure 1.8. For these the interferometer was placed in the light beam immediately after dispersion in a prism spectrometer. Different sections of the circular fringe pattern were thus recorded, due to the different angles of incidence for different wavelengths. Barnes also commented on the increased illumination of the fringes with parallel rather than divergent light.

to have found structure not only in the mercury green line (but quite different from that recorded by Fabry and Perot in 1901) but also, incredibly, in the red cadmium line. If true of course this would have totally undermined not only Fabry and Perot's work but as they noted: '... Il faudrait admettre que des observateurs, comme MM. Michelson, Benoit, Hamy, Chappuis, Macé de Lépinay etc ont travaillé pendant des années sur des longueurs d'onde, sans s'apercevoir qu'ils se servaient d'une unité variable, ou, pour mieux dire, dépourvue de toute signification!' [16]. After further discussion they conclude: '... les composantes que voient MM. Lummer et Gehrke, et qu l'on ne retrouve pas les autres méthodes, seraient analogues aux *fausses images* (que les Américains appellent des *ghost*), données par certains réseaux, et dues, elles aussi, à des variations dans la différence de marche de deux ondes consécutives.' [17]. Obviously the matter must have raised quite a storm at the time. One wonders also about the terminology 'ghost'—was this not also in current usage in England?

A valuable insight into the preoccupations of atomic spectroscopy at this time is provided by the paper of James Barnes, a student of Professors Ames and Wood at Johns Hopkins University. This paper appeared in the *Astrophysical Journal* of 1904 and illustrates the rapid deployment of the Fabry-Perot instrument. With the avowed aim of determining the changes in the components of spectral lines under various conditions he wrote: 'After experimenting a few months with both the Michelson and the Fabry and Perot interferometer, the author was fully convinced that the Fabry and Perot method possessed the advantage for the problem in view, since it shows directly the structure of a given radiation by the simple inspection of the system of fringes. Each fringe is, in fact, a true spectrum of the source . . . '.

Barnes obviously developed a high-class instrument using the base, carriage and screw of a Michelson interferometer made by Gaertner and Co. The glass interferometer plates were of size 3.9 cm by 2.5 cm and about 0.6 cm thick; fine adjustment of parallelism was achieved by change of mercury pressure in the manner similar to that of Fabry and Perot. Adequate stability for very long photographic exposure was obtained '... without the least fear of obtaining a blurred image'. Barnes in fact showed a series of very fine photographs, reproduced as figure 1.8, and also investigated in some detail the effects of defocusing the fringe forming lens. Oddly enough, for such an optically distinguished institution, he uttered (apparently for the first time in print, though regrettably not the last!) the canard that '... a large amount of light is lost, due to reflection from the silvering in contact with plate A . . . , so that only a small percentage is transmitted'. Drawing from this misconception he suggested a reflecting interferometer with the incident beam falling directly onto the second plate and observation of off-axis fringes. As Barnes noted this device bore some relationship to the plate described by Lummer and Gehrke. The observations and measurements of Barnes on many lines in mercury, cadmium, thallium and hydrogen showed a determined effort to get to grips with complex structures clouded by lack of reproducibility of discharge tube and many other uncertainties. He wryly remarked: 'The changes occur so suddenly on the least

change of the surrounding conditions, and sometimes even when no changes apparent to the observer were introduced, . . .’.

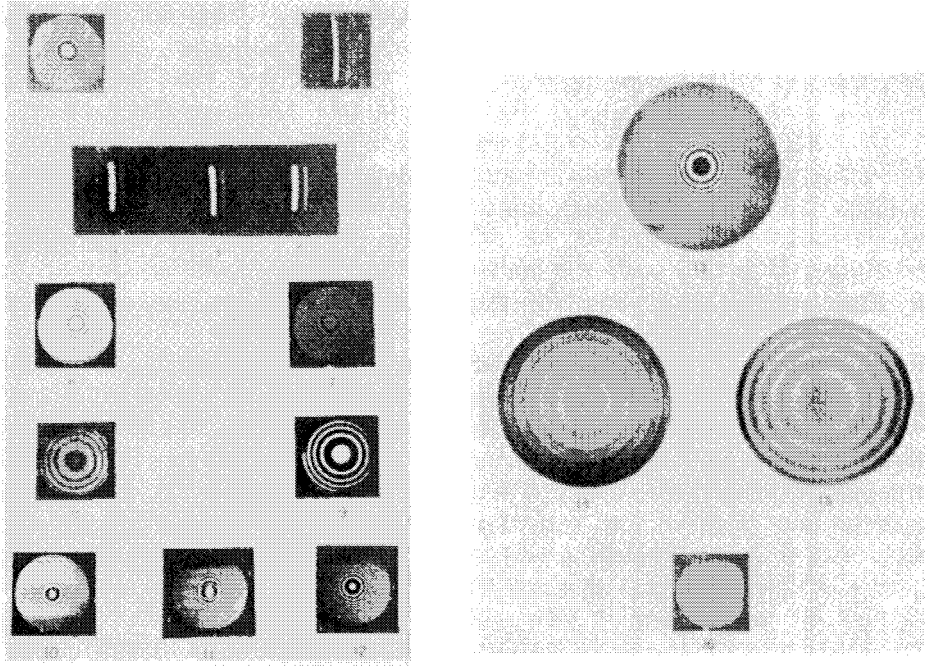


Figure 1.8 Interference fringes photographed by Barnes (1904) under various conditions of plate separation, optical arrangement and focusing of the fringe forming lens. Reprinted courtesy of *The Astrophysical Journal*, published by the University of Chicago Press; © (1904) The American Astronomical Society.

This appears to be the first description and use of the Fabry–Perot interferometer in the United States. A characteristic account of his work in Britain with a Fabry–Perot étalon was given in 1906(a) by Lord Rayleigh, then President of the Royal Society. After noting that ‘. . . it seems to be pretty generally agreed that the use of the diffraction grating . . . must be limited to interpolation . . .’, he rather grandly remarked: ‘. . . this disappointment is partly the result of exaggerated expectations, against which in 1888 I gave what was intended to be a warning’. His second paragraph catches so many nuances of the period that it is worth quoting in full (reproduced by permission of Taylor & Francis Ltd):

Under these circumstances, attention has naturally been directed to interference methods, and especially to that so skillfully worked out by Fabry and Perot. In using an accepted phrase it may be well to say definitely that these methods have no more claim to the title than has the method which employs the grating. The difference between the

grating and the parallel plates of Fabry and Perot is not that the latter depends more upon interference than the former, but that in virtue of simplicity the parallel plates allow of a more accurate construction. In Fabry and Perot's work the wave-lengths are directly compared with the green and red of *cadmium*; and they have obtained numbers, apparently of great accuracy, for artificial lights from vacuum-tubes containing various substances, *e. g.* mercury, for numerous lines from an iron arc, and also for various rays of the solar spectrum. While, so far as I can judge, there has been every disposition to receive with favour work which not only bears the marks of care but is explained with great discrimination, it must still be felt that, in accordance with an almost universal rule, confirmation by other hands is necessary to complete satisfaction. It was with this feeling that about a year ago I commenced some observations of which I now present a preliminary account. I was not without hope that I might be able to introduce some variations which would turn out to be improvements, and which would, at any rate, promote the independence of my results.

Lord Rayleigh distinguished very emphatically between the two forms of mounting: the 'interferometers' of thickness adjustable over a wide range and described as 'elaborate and costly', and the fixed thickness 'étalon' or interference gauge. In his experimental work he employed the latter. For analysis he used a modified procedure, compared with Fabry and Perot, in which P was not determined by auxiliary measurements. Out of a range of possible values the actual fringe order number P was obtained from self-consistency checks made over several wavelengths. Thus rewriting equation (1.7) as

$$\frac{P'}{P} = \frac{\lambda}{\lambda'} \left(1 + \frac{x^2}{8} - \frac{(x')^2}{8} \right) \quad (1.8)$$

where x and x' are the angular diameter of fringes, P' may be calculated from proposed (integral) values of P . The resultant P' should itself be close to an integer and allows selection of P . Unfortunately, for widely differing wavelengths, equation (1.8) requires correction to account for the different effective thickness of the silver films and hence optical thickness of the plate. This of course had been evaluated by Fabry and Perot in their 1899 and 1902 papers. Lord Rayleigh obviously found this correction irksome and rather tetchily remarked on apparent errors and misprints in their discussion.

Lord Rayleigh described his arrangement for viewing the fringes with a telescope and commented on the large adjustment of the eyepiece needed for different colours (although in a footnote: 'I ought perhaps to mention that my eyes have now very little power of accommodation'). To achieve adequate dispersion he used direct vision prisms including a large 20 inch prism made up of glass and carbon disulphide. He described a novel design of discharge tube and in regard to their general reliability and reproducibility remarked darkly: 'There is much in the behaviour of vacuum-tubes which at present defies explanation!'. Using 1 mm and 5 mm étalons Lord Rayleigh made measurements

on lines in cadmium, mercury, zinc, sodium and helium. Where comparison was possible with the results of Fabry and Perot, he found agreement to about one part in a million and often much closer.

In further work described two years later in 1908 Lord Rayleigh employed plate spacers made of Invar ‘in the form of nearly spherical balls, 5 mm in diameter, provided with projecting tongues by which they were firmly fitted to the iron frame . . . The whole mounting was constructed by Mr Enock, and it answered its purpose satisfactorily. There is no doubt, I think, as to the advantage of Invar’. Several measurements were reported including greatly improved ones upon helium. While remarking once more on the complication arising from differing phase changes at the silver surface Lord Rayleigh now found the change between red and green much smaller than his previous measurement and attributed this to tarnish on the earlier silver surfaces. He also described in some detail the construction of a gauge of comparatively large separation. This is shown in figure 1.9; the spacer or ‘distance piece’ consisted of a 30 mm length of glass tubing with three studs or protuberances at each end, whose shaping was ‘. . . effected with a file moistened with turpentine’. This seems to have been crude but effective. Lord Rayleigh discussed several experimental points including the desirability of using an accurately made spacer (to within one wavelength) which required only moderate pressure for adjustment, noting that large pressures gave a less durable state of adjustment. His calculations of the pressure required to compress the spacer a given amount showed good agreement with experiments. In other comments he suggested correction of plate flatness by applying a bending moment between the distance piece and spring pressure, and also figuring inexpensive pieces of plate glass by hydrofluoric acid. In this hazardous process he appears to have achieved some degree of success and noted that this ‘. . . depends upon judgement and practice, and the only general advice that can be given is to make a great many bites at the cherry, and to keep a record of what is done each time . . .’—still admirable advice for the budding spectroscopist in many different contexts!

1.7 Spectroscopic standards

The problem of spectroscopic standards was clearly of increasing importance at this time. In the system of Rowland, absolute wavelength errors amounted to 1 part in 30 000, while relative values showed systematic errors of 1 part in 100 000. Moreover as Fabry and Buisson remarked, Rowland’s measurements were based on the solar spectrum, while metallic arc spectra were usually employed for comparison. These problems were discussed extensively at the first meeting of the International Union for Solar Research. In 1905, at the 2nd Congress held at Oxford, a decision was taken to adopt the wavelengths of a certain number of arc lines as standards. These standards lines were to be established by measurements with the Fabry and Perot method. During the next few years wavelength determinations on the same iron lines were carried out by Fabry and Buisson (1908) at Marseilles, by Eversheim (1907) at the University

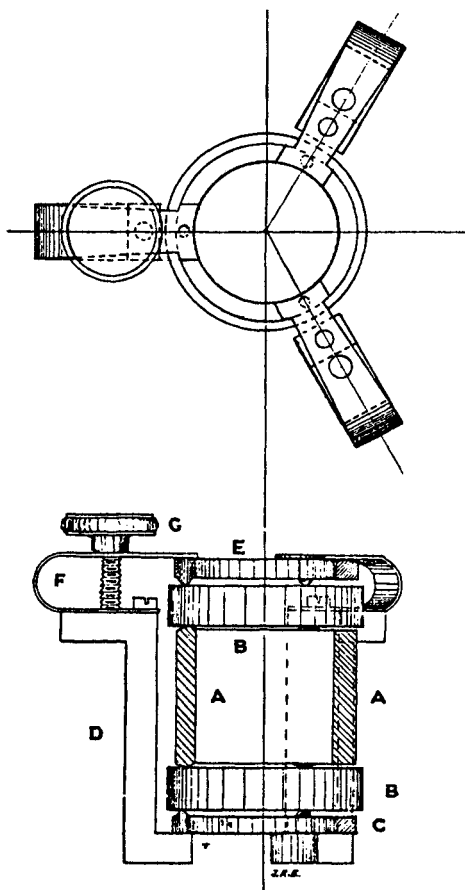


Figure 1.9 The thirty millimetre étalon of Lord Rayleigh (1908) constructed by Mr Enock, showing the plates B and distance piece AA. The lower plate rests upon a brass ring C to which three brass castings D are rigidly attached. The upper ring E is connected with the casting through the spring F. Reproduced by permission of Taylor & Francis Ltd.

of Bonn and by Pfund (1908) at Johns Hopkins University. In 1907 a third meeting took place at Meudon; from the measurements of Benoit, and Fabry and Perot, in good agreement with those of Michelson and Benoit, it became possible to agree on the red cadmium line as the primary standard. From their measurements its wavelength was designated as $6438.4696 \text{ \AA}$, thus providing the definition of the Ångström. Fabry and Buisson (1908) noted that 'it may be considered as certain that it is equal to 10^{-10} meter with an accuracy of about one part in ten million'.

In their report of 1908, appearing in the *Astrophysical Journal* (in the same volume as Pfund's paper) Fabry and Buisson outlined the background to these decisions. They restated many of the basic principles and techniques of

interferometry before describing the photographic equipment they employed to measure 115 different lines. For wavelengths greater than $3600 \text{ \AA}$, glass plates were used, silvered by chemical deposition or cathode sputtering. They noted that the latter technique could produce silver films of ‘remarkable regularity and of very high reflecting power’, but was unfortunately rather inconsistent and unpredictable. Quartz plates were used for shorter wavelengths which were ‘nicked by cathode projection’, the reflectivity of silver being too low in this region.

The complete spectrograph employed by Fabry and Buisson is shown in figure 1.10. The aperture of the fixed spacing interferometer was limited to 8 mm diameter—they noted that ‘thanks to this precaution a slight lack of parallelism of the surfaces was without influence’. Air spacers of 5 mm were used for wavelengths greater than $3600 \text{ \AA}$, and 2.5 mm for less. The plane Rowland grating had a ruled surface $80 \times 50 \text{ mm}$ with 568 lines to the mm and was used in second and third order. The light from the grating passed beneath the reflecting prism onto the photographic plate of size $9 \times 12 \text{ cm}$. The plate holder allowed several exposures to be made on the same plate. Buisson and Fabry in their report provided a wealth of experimental detail and procedure—precalibration checks—lens, grating and plate alignment settings—sequence of photographic recordings etc. For the extreme ultraviolet, because of the risk of fogging the plates by reflection from the objective lens, the collimating arrangement shown in figure 1.10 was abandoned in favour of a separate collimator and

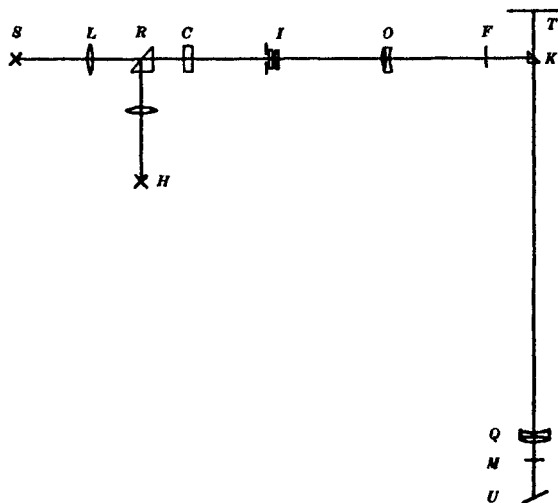


Figure 1.10 The spectrograph employed by Fabry and Buisson in 1908. *S*, *H*, sources; *C*, absorption cell; *I*, interferometer; *F*, focal plane and spectrograph slit; *U*, grating and *T*, photographic plate. This subsequently became a standard arrangement until the advent of photoelectric recording. Reprinted courtesy of *The Astrophysical Journal*, published by the University of Chicago Press; © (1908) The American Astronomical Society.

camera. Several types of photographic plate were used—they remarked that ‘Not until near the end of our research did we try the excellent plates of Wratten and Wainwright, the use of which would have rendered us great service had we known of them earlier’.

Measurements were made by the method of diameters, the mercury green line served as an intermediate standard. Angular calibration was achieved by photographing two ‘metallic landmarks’ of known separation placed across the slit of the spectrograph. The usual corrections for phase change at the metallic films were applied and also those for temperature and pressure of the air. Fabry and Buisson reported a total of 405 measurements employing 43 photographic plates involving 115 lines between 2373 and 6494 Å. The ‘complete example of a measurement’ they gave is shown below (reprinted courtesy of *The Astrophysical Journal*, published by the University of Chicago Press; © (1908) The American Astronomical Society):

Conditions under Which the Negative Was Made

Negative 56. March 17, 1906. Smith plate. Temperature 17°6.
Pressure 765.

Interference apparatus of 5 mm thickness. Glass plates silvered by cathode projection.

Slit-width 0.3 mm.

For the mercury rings, time of exposure 30 seconds.

For the iron rings, time of exposure 10 seconds, wave-length at middle of plate 3800. Inclination of the plate 12°. Intensity of arc current 3 amperes.

Measurement of the Mercury Rings

	1st Ring $P' = 18336$	2d Ring $P' = 18335$	Distance of Standards
Exposure 1	$\delta' = 2.675$	$\delta' = 6.111$	$r' = 8.12$
Exposure 2	$\delta' = 2.735$	$\delta' = 6.115$	$r' = 8.12$

All these lengths are expressed in millimeters.

Angular distance of the standards placed upon the slit, $\theta' = 0.03091$.

The order of interference is then calculated from the formula

$$p' = P' + \frac{P'}{8} \left(\frac{\theta' \delta'}{r'} \right)^2,$$

which gives the following values:

	Calculated from 1st Ring	Calculated from 2d Ring	Mean
Exposure 1	18336.238	18336.242	18336.240
Exposure 2	.248	.244	.246

Final mean, $p' = 18336.243$

The wave-length of the green ray, obtained with these silverings and this interference apparatus, is $\lambda' = 5460.741$.

Consequently $p'\lambda' = 10012.947 \mu$.

Measurement of the Iron Rings

Wave length given by Rowland	3977.891
The same corrected	$\lambda_1 = 3977.76$

Approximate value of the order of interference	$p_1 = \frac{p'\lambda'}{\lambda_1} = 25172.32$	
Angle between standards	$\theta = 0.03040$ radian	
Distance between standards	$r = 8.09$ mm	
Diameter of rings, δ	1st Ring 3.02	2d Ring 5.625
Number of order, P	25172	25171
Computed order of interference	25172.406	25172.407
Mean, p	25172.406	
Rough value of wave-length, $\frac{p'\lambda'}{p}$	3977.7473	
Correction for air	0	
Correction for phase	-0.0019	
Corrected wave-length	3977.7454	

An overall precision of better than 1 part in a million was claimed and good agreement with the Fabry and Perot measurements of 1901. It may be remarked that this paper shows the remarkable maturity of the technique achieved in ten years. The spectrographic arrangement and photographic method, here so admirably described by Fabry and Buisson, remained in widespread use in essentially this form for nearly fifty years. Not until the introduction of photoelectric recording can the technique be said to have changed fundamentally.

The contemporary measurements of other authors deserve comment as they show the wider employment of the method. A preliminary report of wavelength measurements with an étalon, which he described as a 'silvered air-stratum', was made by Eversheim in 1907. (*Astrophysics Journal*, translated from *Z.f.W. Photographie*). In describing the technique of analysis of ring diameter he agreed with Lord Rayleigh in preferring to select integral order numbers from consistency checks for several wavelengths (he suggested five). In his spectroscope the lines to be studied were selected singly by a concave Rowland grating and compared with cadmium green. The glass plates were separated by three steel balls '... figured with an astonishing accuracy', and adjusted for parallelism by spring pressure. However the aperture at the plates was restricted to 3 mm and smaller at the camera lens. Eversheim noted the stringent requirements on temperature stability, one tenth of a degree producing a very perceptible alteration of ring diameter. As an experimental check comparison exposures (to the cadmium source) were made before and after exposure of the line to be measured. With regard to the iron arc source he plaintively commented: 'It is true that such an arc sorely tries the patience of the observer, burning often for a good quarter of an hour without disturbance, whereupon it suddenly makes the wildest leaps and is only with difficulty quieted down'.

Eversheim reported measurements on five iron lines, derived from two different photographs made at different times and different temperatures and taking the three innermost rings, six measures in all. Reasonable agreement with Fabry and Buisson was found although the values were consistently higher by

0.001 to 0.005 Å. Eversheim had no doubt this discrepancy would be found in the displacement of phase due to the silver films, although his own values were closer to those of Fabry and Perot than to Lord Rayleigh's.

In 1908 Pfund reported measurements on 23 iron lines from a good, stable, arc source. The photographic recording he showed is reproduced as figure 1.11. The étalon spacers were made of three Invar studs equally spaced in a brass ring. Final length adjustment to make the studs equal within a quarter-wavelength was carried out by rubbing them lightly against a smooth piece of wood or paper. Pfund improved the reflectivity of his chemically silvered films by buffing them with rouge on a 'toilet powder puff consisting of innumerable fine feathers'. This apparently removed without scratching the thin, pale yellow surface film on the silver and more than doubled the number of reflections of a gas flame viewed through the two mirrors. The temperature controlled étalon assembly is shown in figure 1.12 and was sufficiently effective that fringe diameters did not change perceptibly over periods of several hours. After the usual phase change correction, Pfund's measured values nowhere differed by more than 0.004 Å from those of Fabry and Buisson. In an interesting comment on accuracy he reported that successive readings of ring diameters might be repeated with an error of 1 part in 500 or 600. However delayed measurements and those on different rings suggested that the highest accuracy that could be claimed was in fact little better than 1 part in 150, corresponding to about 0.004 Å.

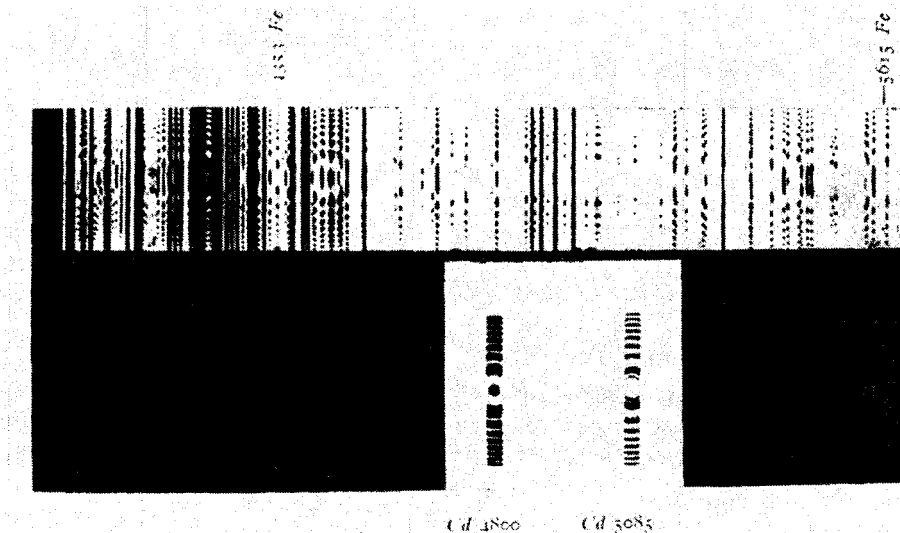


Figure 1.11 A particularly fine recording of iron and cadmium spectra shown by Pfund (1908). For cadmium the spectrograph slit width was increased by a factor of 2.5. Reprinted courtesy of *The Astrophysical Journal*, published by the University of Chicago Press; © (1908) The American Astronomical Society.

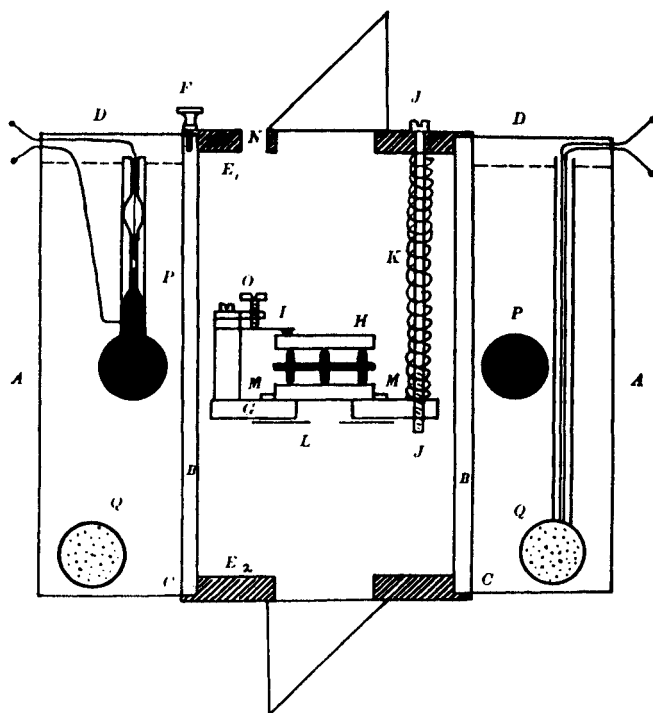


Figure 1.12 The horizontally mounted étalon assembly used by Pfund (1908). The étalon assembly was supported inside the brass tube *B* by three adjustable brass rods *J*. The temperature was kept constant with the water bath *A*, heaters at *Q* and circular temperature regulator *P* filled with mercury. Reprinted courtesy of *The Astrophysical Journal*, published by the University of Chicago Press; © (1908) The American Astronomical Society.

1.8 Application to the study of nebulae

With his lifelong interest in astronomy, Fabry soon sought to extend interference methods beyond the study of solar absorption lines. In 1911 he and Buisson published a preliminary note in the *Astrophysical Journal* on the application to nebulae whose spectra consist of a small number of bright lines. They wrote that ‘... It is desirable that there should be no mixing of the radiations emitted by the different parts of the object, and consequently that the sharp image of the nebula should be formed in the same plane as that of the rings. ... The interference apparatus is placed at the end of the telescope set upon the nebula for an eye focused for infinity ... on looking through this, if the eye is focused for infinity it will see a sharp image of the nebula on which the interference fringes will be projected. ... For photographic observation the light ... is collected by an objective of short focus in the focal plane of which are superposed the real image of the nebula and that of the ring’.

For their experimental work they were fortunate enough to employ the

equatorial of the Observatory of the University of Marseilles. Interference rings due to $\lambda 5007$ were observed visually, with étalon spacers up to 5 mm, from the nebula in *Orion*. Photographic recordings of $H\gamma$ at $\lambda 4341$ and an unknown line at $\lambda 3727$ were obtained with a spacer of 0.6 mm and an exposure of 75 min. In concluding this initial report they noted some very important applications: 'It would be easy . . . to derive the radial velocity of the object in using the hydrogen lines. The variations of the wavelength from one point to another would give us the circulatory movement of the gas . . . an indication as to the temperature would be furnished by the size of the hydrogen lines, while the size of the lines of unknown origin would give us a clue to the atomic weight of the gas which forms them'. In 1914 a more detailed account of their work was published in collaboration with H Bourget, Director of the Observatory. The arrangement for photographing the fringes is shown in figure 1.13. Spacers between 0.1 and 3 mm were employed, the interferometer plates were 4 cm diameter and 1 cm thick; one pair of plates were silvered to give a transmission of 0.15 and reflectivity 0.74 (at Hg $\lambda 5461$) and a second pair to give 0.30 and 0.60 respectively. These were obtained by 'cathodic projection, following a method which permitted us to obtain at once exactly and very rapidly the desired thickness of silver'. In a footnote they noted their intention to publish a description of this technique and details of the construction of the étalons. This does not seem to have appeared, probably due to the impact of the Great War.

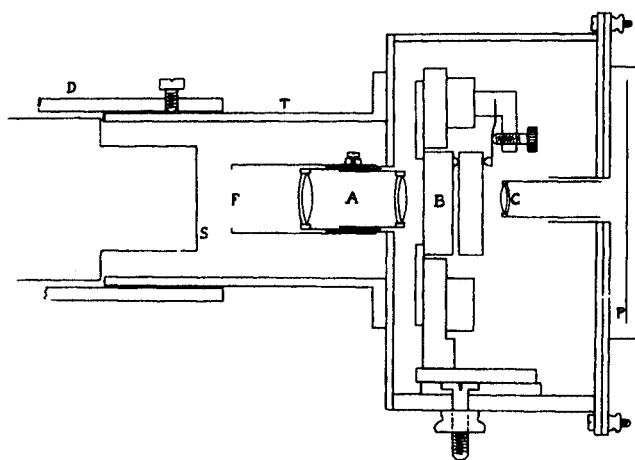


Figure 1.13 The optical arrangement mounted on a telescope: F, focal plane of telescope; A, collimating arrangement; B, étalon plates; C, fringe forming objective; P, focal plane and photographic plate. (From Buisson *et al* 1914, figure 1.) Reprinted courtesy of *The Astrophysical Journal*, published by the University of Chicago Press; © (1914) The American Astronomical Society.

The equipment shown in figure 1.13 weighed 4.5 kg; it was mounted on the large reflecting telescope at Marseilles whose 80 cm diameter glass mirror had been figured by Foucault in 1862. At the photographic plate an angular

magnification was achieved 80 times larger than if the objective C were pointed directly at the sky. Absorption filters were used to isolate different lines; a list of photographic exposures is reproduced below (reprinted courtesy of *The*

TABLE I
LIST OF PLATES

Date 1914	Radiation Used	Thickness	Blocks	Exposure
January 27.....	H _γ +H _β	1 mm	Steel	1½ hours
28.....	H _γ +H _β	2 mm	"	"
30.....	H _γ	1 mm	"	1 hour
31.....	3728	1 mm	"	1½ hours
February 2.....	"	110 μ	Mica	" "
13.....	"	130 μ	Wire	" "
14.....	H _γ	1 mm	Invar	" "
16.....	3728	1 mm	"	1½ hours
28.....	H _γ	2 mm	"	2 "
March 2.....	3728	0.64 mm	"	" "
3.....	"	2 mm	"	" "
5.....	"	2.8 mm	"	" "
7.....	H _γ	2 mm	"	" "
12.....	H _γ	1 mm	"	" "

Astrophysical Journal, published by the University of Chicago Press; © (1914) The American Astronomical Society). The photograph they showed is given in figure 1.14. Since the rings may be distorted by differences of radial velocity from point to point, the exact centre, defined by the foot of the normal to the silvered surfaces, needs to be found. Fabry and his colleagues described their technique (aided by the cross-wire image shown in figure 1.14) with characteristic thoroughness. From their measurements on H_γ for 58 points distributed in 12 directions, they deduced that the nebula was receding from the sun with an average value of 15.8 km s⁻¹. They noted that this was in satisfactory agreement with earlier values and described the variations they perceived, amounting to 10 km s⁻¹ over quite small areas, with a rotary movement about a line southeast-northwest. For the temperature of the gas they suggested an upper value estimate of 15000 degrees. In conclusion they emphasized: ‘... the simplicity of the apparatus used and the ease with which it can be mounted on the telescope ... the interferential apparatus does not cause the loss of much light and permits the study of objects of very feeble intrinsic brightness’.

1.9 Line broadening, Doppler and gravitational shift, and other studies

Measurement of absorption lines was discussed in detail by Fabry and Buisson in 1910(a), several years after their first work on solar spectra. In particular they considered the problem of correctly matching an interferometer to the spectro-scope that provides auxiliary dispersion. As a good working rule they suggested that the interferometer spacing e should be chosen to given an order of interference p (equal to $2e/\lambda$) lying between $\lambda/20\delta\lambda$ and $\lambda/\delta\lambda$ where $\delta\lambda$ is the width of the absorption line. The slit width is supposed chosen equal to $\sim \lambda/p$,

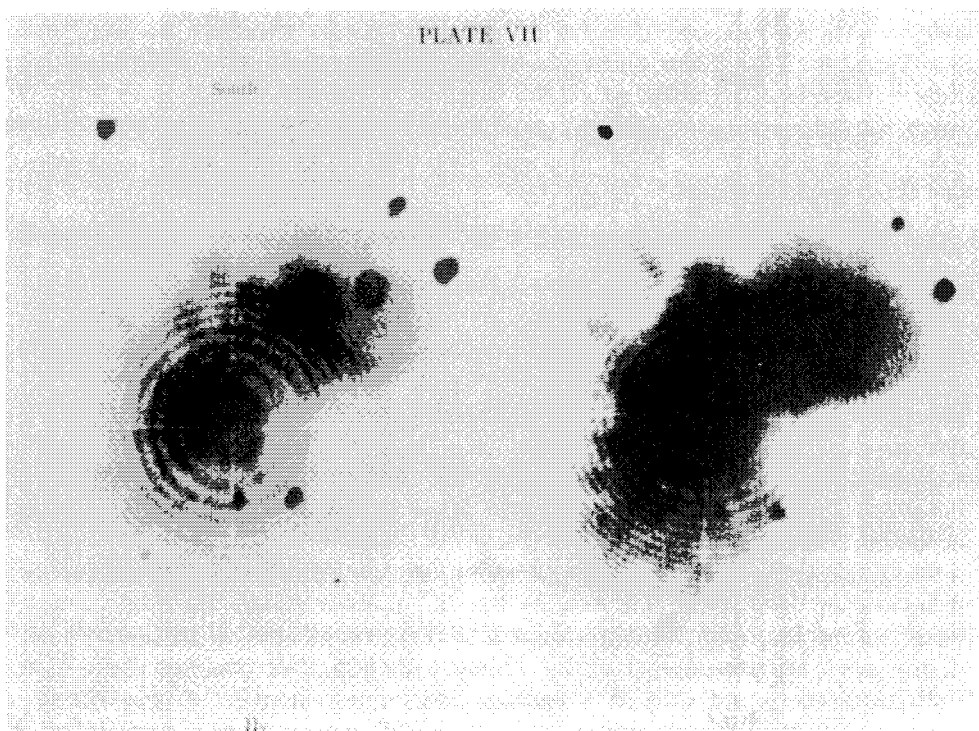


Figure 1.14 Interference photographs of the Orion Nebula taken in 1914 by Buisson *et al* (plate VII) using the reflecting telescope of the Observatory of Marseille. Reprinted courtesy of *The Astrophysical Journal*, published by the University of Chicago Press; © (1914) The American Astronomical Society.

the range of the interferometer, so that the channel spectra due to a continuous source are just rendered unobservable. (The slit thus lies between $20\delta\lambda$ and λ .)

Another instrumental analysis of interferometers with intrinsic dispersion in the optical delay path was given in 1908 by von Baeyer. This was of particular relevance to the Lummer plate where the successive beams are delayed within the dispersive glass plate. For the air-spaced Fabry–Perot interferometer the effect is of little consequence. Many years later Kastler (1962) commented on the problem for light sources within a cavity. Recently Gardner and Vaughan (1982) rederived the analysis in order to explain the transmission of a cavity containing a cell of calcium vapour (Appendix 7).

In 1912 Buisson and Fabry presented interference measurements on the width of spectral lines from helium, neon and krypton. In passing they compared their own interferometer with other multiple-beam instruments and noted: ‘... on obtient la véritable composition des raies complexes, et l’on voit chaque satellite à sa place comme dans un spectroscopie. Cette particularité est commune à tous les appareils à ondes multiples. Il en a été inventé plusieurs autres (échelon de Michelson, lame de Lummer); mais le pouvoir de définition de ces appareils est fixe et ne peut pas s’adapter à la finesse des phénomènes étudiés; d’autre part,

leur construction est très difficile, ce qui peut rendre illusaires les avantages théoriques . . . [18]. Their technique of line width analysis relied on finding the order of interference N for which the interference fringes ceased to be visible, which they maintained could be established very exactly. For lines broadened according to gas kinetic theory and the Doppler-Fizeau effect they calculated that this would occur for

$$N = 1.22 \times 10^6 \sqrt{m/T} \quad (1.9)$$

where m was the atomic mass and T the temperature. For cooling with liquid air compared with room temperature they found a line narrowing almost as great as expected—the order number N increasing by 1.66 for helium, 1.60 for neon and 1.58 for krypton, compared with a theoretical 1.73. As they noted: ‘Un léger excès de la température du gaz sur celle du bain expliquerait la petite différence entre le nombre théorique et le résultat des mesures’ [19]. Other sources of line width discussed by Fabry and Buisson included a basic collision theory of line broadening, following Shönrock, which they suggested would be negligible for the low gas pressures (of order 1 mm) employed in their Geissler tubes. Similarly broadening effects of self-absorption within such discharges were dismissed; in both cases one now knows (§7.4) some broadening could occur. While noting that many lines were complex they commented on the narrowness of the green krypton line at $\lambda 5570$ which, with liquid air cooling, had a line width of $0.006 \text{ \AA}$ and showed a limiting interference number of $N = 950\,000$, corresponding to an interferometer spacing of 26.5 cm.

Potential applications to atomic spectroscopy and astronomy, particularly deriving from line width studies and the measurement of T and m , were discussed in some detail by Fabry and Buisson. In hydrogen they showed that lines in the second spectrum had a width corresponding to $m = 1$ and thus could not be attributed to impurities or an association of several atoms as had been suggested. In an electric arc between iron electrodes, run at low pressure, a temperature of 2400°C was deduced; the line width increased rapidly at atmospheric pressure, this being attributed to collisional effects. Flame temperatures of 2000°C were deduced. The width of so-called ‘enhanced lines’ was also studied. For these Lockyer had suggested: ‘sa célèbre hypothèse’ of proto elements—a disassociation of atoms into lighter atoms. For cadmium and mercury Fabry and Buisson showed that such enhanced lines were no broader than the ordinary lines, and did not necessarily require high temperatures for their generation but rather high electric field strength.

The application of these methods to the width and origin of spectral lines was discussed in 1915 by both Lord Rayleigh and Thomas Merton, a lecturer in spectroscopy at King’s College, London. In expression (1.9) Lord Rayleigh had proposed a numerical value of 1.427×10^6 ; however Merton, from observations on helium and neon spectra in vacuum tubes, obtained experimental values very close to that proposed by Fabry and Buisson— 1.22×10^6 . Merton employed a ‘Fabry and Perot sliding interferometer’ but otherwise gives little detail of the design or manufacture. Both photographic and visual measurements were made on lines in calcium, strontium, barium, argon and helium. Merton’s paper

provides a fascinating insight into the problems of the day as spectroscopists struggled with the interpretation and assignment of band spectra, sharp and diffuse series etc, all the while confused by uncertainty and non-reproducibility of flame, discharge and arc sources. Merton at least attempted most conscientiously to describe precise conditions of excitation—pressure, current, region of observation, cooling etc. In hindsight one can see how troublesome broadening due for example to self-absorption, isotope shifts and electric field effects must have been. It is in fact remarkable how much information could be extracted, although in many instances all that could be done, as Merton remarked, was ‘... to admit to the existence of some hypothetical cause of broadening in this case’.

The narrowness of krypton lines and their advantages for metrology was commented on again in 1913 by Buisson and Fabry. They reported measurements of wavelength employing a 100 mm étalon which had been made by M Jobin ‘... pour la Chambre des Poids et Mesures de l’Empire de Russie’. Superposition fringes in white light were used for comparison with a 25 mm spacer; the green and yellow lines were measured against the cadmium red line with a precision better than 1 in 10^8 .

In the following year Fabry and Buisson described the first precision laboratory verification of the Doppler–Fizeau principle. For this they illuminated a rapidly rotating paper disc with a mercury lamp. Opposite extremities of the perpendicular diameter were then successively viewed through an interferometer of 65 mm spacing. A rotational speed of 200 rps gave a speed at the circumference of 100 ms^{-1} and a shift of $\frac{1}{3}$ of a fringe across the diameter for violet light. They commented that ‘Une seule expérience, faite avec peu de précautions, a donné un résultat concordant à 2 pour 100 près avec le résultat calculé’ [20].

The active and continued scientific interests of Perot are evidenced by his attempts in 1921 to verify the relativity principle in solar lines of magnesium. A relative gravitational red shift of 2×10^{-6} was predicted; Perot and Lindstedt in 1911 had measured shifts, between solar wavelengths and an arc at atmospheric pressure, of $6 \times 10^{-3} \text{ Å}$ or 1.15×10^{-6} . The question arose as to what correction was required for the pressure in the solar atmosphere where the lines originated. By ingenious comparison of the pressure dependence of a group of magnesium lines in an arc source (measured interferometrically), Perot was able to reanalyze his original photographs of 1911, which suggested that the solar pressure was in fact low. Applying an appropriate correction Perot obtained a relative difference between the solar and terrestrial lines of 2.5×10^{-6} , in reasonable agreement with Einstein’s prediction.

This study apparently inspired Buisson and Fabry (1921) to re-examine their results of 1909 and 1910 on iron lines in the solar spectrum. These were well known to show a shift towards the red. This had been previously attributed to pressure effects, although as they now remarked this was unsatisfactory because of various anomalies of observed width and shift. Under the assumptions of low solar pressure and negligible pressure shift the mean displacement for 32 solar lines was within 20% of Einstein’s theory and thus supported the hypothesis that ‘L’effet Enstein est la seule cause de déplacement des raies du spectre solaire’.

However they cautiously added that ‘Dans toutes les comparaisons de ce genre, il faut se tenir soigneusement en garde contre les phénomènes accessoires de l’arc qui peuvent donner aux raies une fausse position’ [21].

1.10 Conclusion

By the early 1920s the Fabry–Perot interferometer had become a well established spectroscopic equipment. Figure 1.15 shows an example of a particularly robust and easily adjusted form of interferometer due to Goos (1912), which also permitted fairly rapid change of plate separation. Two early reviews of the design and applications of the instrument may be found in the *Handbuch der Physikalischen Optik* (1927) in articles by Feussner and Janicki, and by Kösters. Figure 1.16 shows a fixed interferometer of the period illustrated by Kösters with rectangular plates in which the spacer was a U channel made of Invar with three projecting studs on the end faces. Another early instrument from the first decades of the century, made by A Hilger Ltd of London is shown in figure 1.17 and was illustrated by Candler. Writing in 1951 he commented that this variable gap type of interferometer had become obsolete since even the best instrument maker ‘cannot work ways which will keep the two plates parallel with sufficient accuracy’.

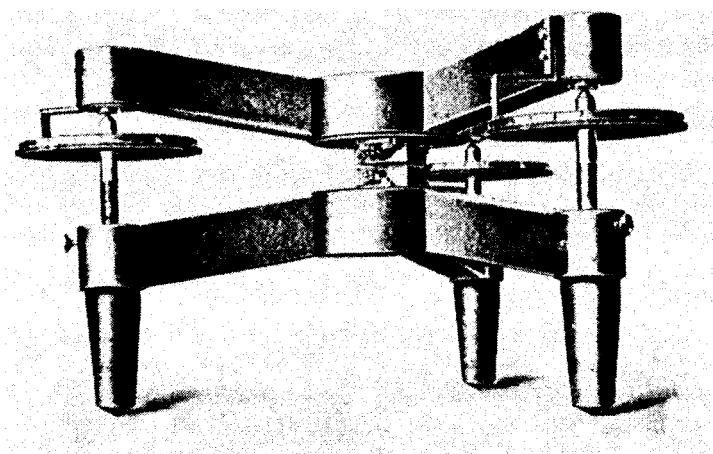


Figure 1.15 A horizontally mounted interferometer described by Goos (1912) working in Hamburg. Reproduced by permission of DFVLR.

It is perhaps appropriate to end this chapter with a mention of Fabry’s monograph ‘Les Applications des Interférences Lumineuses’ published in 1923. Fabry was now Directeur Général de L’Institut d’Optique Théorique et Appliquée and Professor at the Sorbonne. In his first chapter he described all manner of interferences in optics before considering the application to the measurement

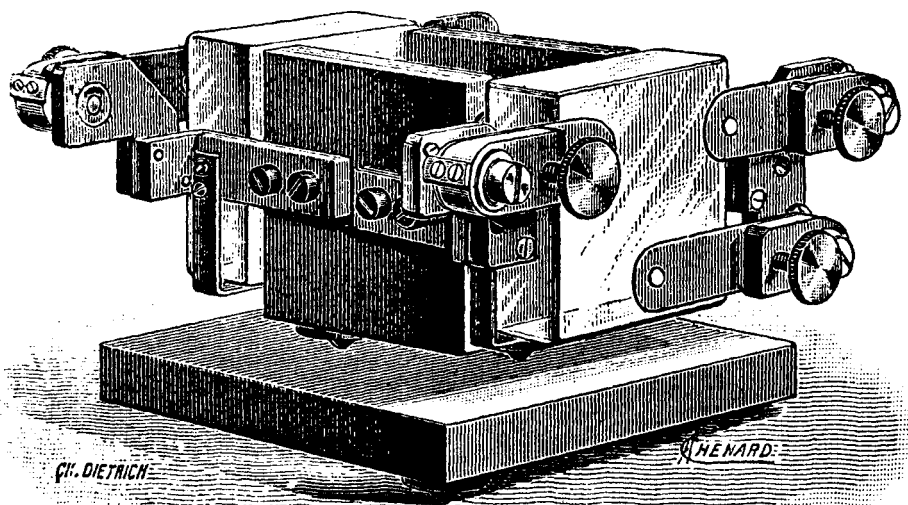


Figure 1.16 An étalon shown by Kösters (1927, figure 230). The spacer is an Invar U channel.

of small displacements and deformations, and in metrology. A short third chapter was devoted to measurement of small angles, before a full account of applications to spectroscopy in the fourth chapter. As might be expected the Fabry-Perot interferometer was described at length, and the text abounds with that characteristic blend of theoretical explanation, and experimental and technical detail, in which Fabry obviously took such delight, and which must have proved invaluable to his readers.

Some idea of the increased use of Fabry-Perot interferometers is given by the list of leading centres involved with measurement of standard wavelengths. These included not only Marseilles, Pfund at Baltimore and Eversheim at Bonn, already mentioned, but also Burns at Washington, Goos at Hamburg and Saint-John at Pasadena. For the future Fabry suggested that, apart from astrophysical measurements, valuable studies would be found in isotope shifts (citing recent work of Merton (1920) on lead), line broadening and shift due to source conditions, the structure of complex lines, and the Zeeman effect. These topics have remained of interest to the present and, for the most part, the basic manner of study stayed essentially unchanged from Fabry and Perot's day until the invention of the laser. Where appropriate the history of those modifications and extensions of technique made during the succeeding decades are outlined in the following chapters. Apart from photoelectric recording however, it is difficult to point to anything that had not been at least envisaged by Fabry and Perot in their early work. This perhaps may stand as the highest possible tribute to their endeavours.

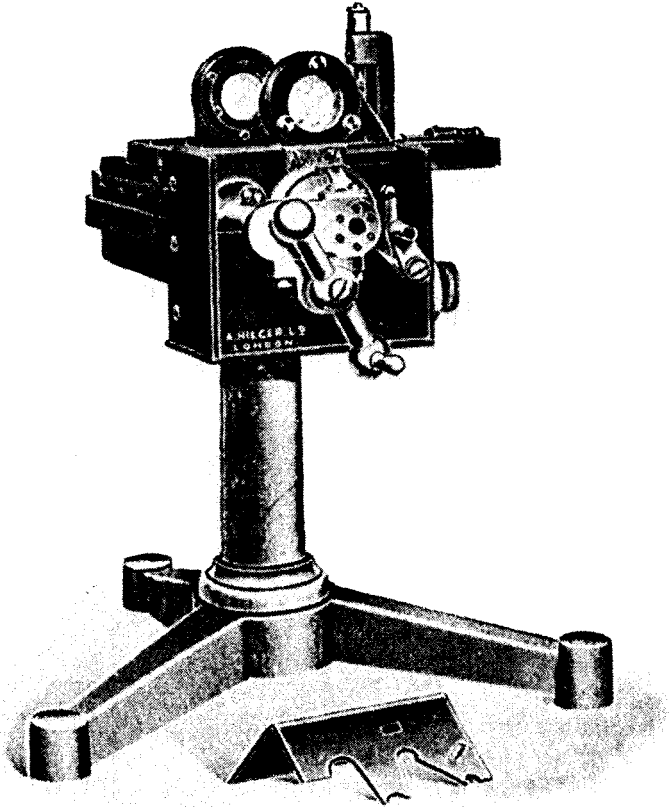


Figure 1.17 An interferometer with one movable plate from the first decades of the century (shown by Candler 1951).

Translation of passages from Chapter 1

[1] It is possible to see simultaneously the two systems of fringes if one arranges them in the appropriate conditions provided by study of Airy's formulae.

[2] . . . it is easy to see that $I + I'$ show two identical maxima separated by minima little different from one another but never zero; the number of fringes will be doubled.

One obtains identical results for the transmitted light since the transmitted rings are complementary to the reflected rings.

[3] . . . it suffices to produce rings between two transparent plates coated with a silver film sufficiently thin such that the whole is slightly transparent for sodium light; after aligning the two surfaces, and for a suitable distance between the plates, the doubled rings appear very clearly.

[4] The *doubling* of fringes provided by sodium light has already been described by M Boulouch (1); they replace the disappearances discovered by Fizeau in the case of ordinary thin plates.

[5] On the fringes of thin silvered plates and their application to the measurement of small thicknesses of air.

[6] . . . have led to the study of optical phenomena interesting in themselves, and to the invention of a method of measurement of small thicknesses which will doubtless be useful in other cases.

[7] This factor can be considered as defining the fineness of the fringes.

[8] . . . each fringe describes a line of equal spacing in the thin layer. Consequently their shape depends absolutely on the shape of the surface . . .

[9] One can, just as for fringes in ordinary thin plates, see them in the case where the surfaces are exactly flat and parallel, by scanning across the system with a lens or eye accommodated for infinity. The fringes which always look like narrow lines, appear then in the form of rings, *localized at infinity*, whose centre is in the direction normal to the surfaces which limit the thin layer. The diameters of these rings reduce in proportion as the thickness of the layer that produces them increases.

[10] Theory and application of a new method of interference spectroscopy. Interference methods for the measurements of large spacings and comparison of wavelengths.

[11] The dispersive power of this small equipment is consequently of the same order as that of the best prism or grating spectroscopes.

[12] . . . one can thus estimate variations of thickness of less than $\frac{1}{20}$ th of a micrometre.

[13] it is not possible to speak of the spectrum of a substance without specifying the conditions of production.

[14] Spectroscopists of the American school make frequent use of this source, in France it is much less known and deserves to be more employed . . .

[15] I placed for this, in the focal plane where the fringes are sharpest, the slit of an astigmatism-free spectroscope. On opening this slit a small amount one sees, in the eyepiece of the spectroscope, each line in the form of a rectangle in which the corresponding rings stand out. . . .

In all cases one has a general view of the spectrum, and one cannot commit the slightest error in examination of the radiation. All the rings are, at one and the same time, concentrated in the corresponding small rectangle, and a single photographic exposure allows them to be obtained all together. Thus the light is much better used than in the former procedure.

[16] . . . one would have to admit that experimentalists such as MM. Michelson, Benoit, Hamy, Chappuis, Mace de Lépinay etc had worked for several years on wavelengths without noticing that they were dealing with a spurious unity, or to speak bluntly, one which was devoid of any significance!

[17] . . . the components seen by MM. Lummer and Gehrke, and which are not found by any other methods, are analogous to *false images* (which the Americans call *ghosts*) given by certain gratings, and due, amongst other things, to the variation in path difference between two consecutive waves.

[18] . . . one obtains the true composition of complex lines, and one sees each satellite in its place as in a spectroscope. This specific property is common to all multiple-beam equipment. It has been invented in several other ways (the échelon of Michelson and Lummer's plate); but the power of definition of these equipments is fixed and cannot be adapted to the line width of the phenomenon under study; furthermore their construction is very difficult and renders illusory any theoretical advantages . . .

[19] A slight excess in the temperature of the gas above that of the bath would explain the small difference between the theoretical figure and the experimental result.

[20] A single trial, made with few precautions, has given a result agreeing within 2 parts in 100 with the calculated value.

[21] The Einstein effect is the only cause of displacement of the solar spectrum. . . . In all comparisons of this type one must painstakingly guard against additional phenomena in the arc which can give the lines a false position.

Historical references to 1927 in chronological order

This is by no means an exhaustive list of Fabry and Perot's papers or of contemporary spectroscopic work. A list of over one hundred and forty early references may be found for example in Fabry's monograph 'Les Applications des Interférences Lumineuses' published in 1923. (Note: for the remaining chapters all references, including a few of early date, are in alphabetical order at the end of the volume.)

Airy G B 1831 *Mathematical Tracts on the Lunar and Planetary Theories, The Figure of the Earth, Precession and Nutation, The Calculus of Variations and The Undulatory Theory of Optics* 2nd edn (Cambridge: J Smith, Printer to the University, for J and J J Deighton)

Airy G B 1833 On the phaenomena of Newton's rings when formed between two transparent substances of different refractive powers *Phil. Mag.* **2** 20–30

Haidinger W 1849 Ueber die schwarzen und gelben Parallel-Linien am Glimmer *Poggendorff's Annal. (Lpz)* **LXXVII** 219–28

Haidinger W 1855 Die Interferenzlinien am Glimmer. Berührungsringe und Plattenringe *Poggendorff's Annal. (Lpz)* **XCVI** 453–68

Boulouch M R 1893 Dédoublément des franges d'interférence en lumière naturelle *J. Physique* **2** 316–20

Fabry Ch and Perot A 1897 Sur les franges des lames minces argentées et leur application à la mesure de petites épaisseurs d'air *Ann. Chim. Phys.* **12** 459–501

Edser E and Butler C P 1898 A simple method of reducing prismatic spectra *Phil. Mag.* **46** 207–16

Perot A and Fabry Ch 1899a Sur l'application de phénomènes d'interférence à la solution de divers problèmes de spectroscopie et de métrologie *Bull. Astronomique* **16** 5–32 (Engl. Transl. *Astrophys. J.* **9** 87–115)

- Fabry Ch and Perot A 1899 Théorie et applications d'une nouvelle méthode de spectroscopie interférentielle *Ann. Chim. Phys.* **16** 115–44
- Perot A and Fabry Ch 1899b Méthodes interférentielles pour la mesure des grandes épaisseurs et la comparaison des longueurs d'onde *Ann. Chim. Phys.* **16** 289–338
- Fabry Ch, Macé de Lépinay J and Perot A 1899a Mesure en longueurs d'onde d'un cube de quartz de 4 cm de côté *C. R. Acad. Sci., Paris* **128**
- Fabry Ch, Macé de Lépinay J and Perot A 1899b Sur la masse du décimètre cube d'eau *C. R. Acad. Sci., Paris* **129**
- Fabry Ch and Perot A 1900a Sur les sources de lumière monochromatiques *J. Physique* **9** 369–82
- Fabry Ch and Perot A 1900b Nouvelle source de lumière pour la spectrométrie de précision *C. R. Acad. Sci., Paris* **130** 406–9
- Perot A and Fabry Ch 1900 Détermination de nouveaux points de repère dans le spectre *C. R. Acad. Sci., Paris* **130** 492–5
- Fabry Ch and Perot A 1901 Sur un nouveau modèle d'interféromètre *Ann. Chim. Phys.* **22** (Engl. Transl. *Astrophys. J.* **13** 265–72, PLATE IX)
- Perot A and Fabry Ch 1901 Mesure en longueurs d'onde de quelques étalons de longueur a bouts *Ann. Chim. Phys.* **24** 119–39
- Fabry Ch and Perot A 1902 Mesures de longueurs d'onde en valeur absolue, spectre solaire et spectre du fer *Ann. Chim. Phys.* **25** (Engl. Transl. *Astrophys. J.* **15** 73–96, 261–73)
- Lummer O and Gehrke L 1903 Ueber die Anwendung der Interferenzen an planparallelen Platten zur Analyse feinsten Spektrallinien *Drude's Annalen* **X** 457
- Perot A and Fabry Ch 1904 Sur la séparation des raies spectrales très voisines a propos d'un travail récent de MM. Lummer et Gehrke *J. Physique* **3** 28–32
- Barnes J 1904 On the analysis of bright spectrum lines *Astrophys. J.* **19** 190–211
- Fabry Ch 1904 On the wavelength of the cadmium line at $\lambda 5086$ *Astrophys. J.* **19** 116–18
- Fabry Ch and Perot A 1904 On the corrections to Rowland's wavelengths *Astrophys. J.* **19** 119–20
- Fabry Ch 1905 Sur un nouveau dispositif pour l'emploi des méthodes de spectroscopie interférentielle *C. R. Acad. Sci., Paris* **140** 848–51
- Hamy M 1906 Sur les franges de réflexion des lames argentées *J. Physique* **5** 789
- Lord Rayleigh 1906a Some measurements of wave-lengths with a modified apparatus *Phil. Mag.* **11** 685–703
- Lord Rayleigh 1906b On the interference-rings, described by Haidinger, observable by means of plates whose surfaces are absolutely parallel *Phil. Mag.* **12** 489–93
- Eversheim P 1907 Determination of wave-lengths of light for the establishment of a standard system *Astrophys. J.* **26** 172–90 (translated from *Z. fur Wissensch. Photographie* **5** 152–80)
- Lord Rayleigh 1908 Further measurements of wave-lengths and miscellaneous notes on Fabry and Perot's apparatus *Phil. Mag.* **15** 548–58
- von Baeyer O 1908 Über den Einfluss der Dispersion auf den Gangunterschied beim Interferenzspektroskop nach Lummer-Gehrcke *Phys. Zeitschr.* **9** 831–4
- Fabry Ch and Buisson H 1908 (translated by R R Tatnall) Wave-length measurements for the establishment of a system of spectroscopic standards *Astrophys. J.* **28** 169–96

- Pfund A H 1908 A redetermination of the wave-lengths of standard iron lines *Astrophys. J.* **28** 197–211
- Fabry Ch and Buisson H 1909 Comparaison des raies du spectre de l'arc électrique et du soleil. Pression de la couche renversante de l'atmosphère solaire *C. R. Acad. Sci., Paris* **148** 688–90
- Fabry Ch and Buisson H 1910a Interférences produites par les raies noires du spectre solaire *J. Physique* **9** 197–205
- Fabry Ch and Buisson H 1910b Application de la méthode interférentielle à la mesure de très petits déplacements de raies. Comparaison du spectre solaire avec le spectre d'arc du fer. Comparaison du centre et du bord du soleil *Astrophys. J.* **31** 97–119
- Buisson H and Fabry Ch 1910 Mesures de petites variations de longueurs d'onde par la méthode interférentielle. Application à différents problèmes de spectroscopie solaire *J. Physique* **9** 298–316
- Eversheim P 1910 Measurements of wave-lengths of standard iron lines *Astrophys. J.* **31** 76–7
- Perot A and Mlle Lindstedt 1911 Sur la longueur d'onde de la raie solaire b_2 *C. R. Acad. Sci., Paris* **152** 1367–70
- Fabry Ch and Buisson H 1911 Application of the interference method to the study of nebulae *Astrophys. J.* **33** 406–9
- Goos F 1912 Eine neue Form des Fabry-Perotschen Interferometers mit veränderlichem Plattenabstand *Z. Instrum.* **32** 326–8
- Perot A 1912 Sur le mouvement apparent des vapeurs dans l'atmosphère solaire *C. R. Acad. Sci., Paris* **154** 1684–6
- Buisson H and Fabry Ch 1912 La largeur des raies spectrales et la théorie cinétique des gaz *J. Physique* **2** 442–64
- Benoit J R, Fabry C and Perot A 1913 Nouvelle détermination du rapport des longueurs d'onde fondamentales avec l'unité métrique *Trav. Bur. Int. Pds. Mes.* **15** 3
- Buisson H and Fabry Ch 1913 Sur les longueurs d'onde des raies du krypton *C. R. Acad. Sci., Paris* **156** 945–7
- Fabry Ch and Buisson H 1914 Vérification expérimentale du principe de Doppler - Fizeau *C. R. Acad. Sci., Paris* **158** 1498–9
- Buisson H, Fabry Ch and Bourget H 1914 An application of interference to the study of the orion nebula *Astrophys. J.* **40** 241–58
- Merton T R 1915 On the application of interference methods to the study of the origin of certain spectrum lines *Proc. R. Soc. A* **91** 421–31
- Lord Rayleigh 1915 On the widening of spectrum lines *Phil. Mag.* **29** 274–84
- Fabry Ch and Buisson H 1919 Description et emploi d'un nouveau microphotomètre *J. Physique* **9** 37–46
- Merton T R 1920 On the secondary spectrum of hydrogen *Proc. R. Soc. A* **96** 382–8; On the spectra of isotopes *Proc. R. Soc. A* **96** 388–95
- Buisson H and Fabry Ch 1921 Sur le déplacement des raies solaires sous l'action du champ de gravitation *C. R. Acad. Sci., Paris* **172** 1020–2
- Perot A 1921 Mesure de la pression de l'atmosphère solaire dans la couche du magnésium et vérification du principe de relativité *C. R. Acad. Sci., Paris* **172** 578–81

- St John Ch E and Babcock H D 1921 Wave lengths of lines in the iron arc from grating and interferometer measures $\lambda 3370\text{--}\lambda 6750$ *Astrophys. J.* **53** 260–99
- Fabry Ch 1923 Les Applications des Interférences Lumineuses *Éditions de la Revue d'Optique théorique et instrumentale* (Paris)
- Feussner W and Janicki L 1927 Interferenzerscheinungen dünner Blattchen: 10. Interferometer von Fabry und Perot in *Handbuch der Physikalischen Optik* ed. E Gehrcke (Leipzig: Barth) pp 430–3
- Kösters W 1927 Anwendung der Interferenzen zu Messwecken in *Handbuch der Physikalischen Optik* ed. E Gehrcke (Leipzig: Barth) pp 476–85

Historical Background

- Airy G. B. 1831 *Mathematical Tracts on the Lunar and Planetary Theories, The Figure of the Earth, Precession and Nutation, The Calculus of Variations and The Undulatory Theory of Optics* 2nd edn (Cambridge: J Smith, Printer to the University, for J and J J Deighton)
- Airy G. B. 1833 On the phaenomena of Newton's rings when formed between two transparent substances of different refractive powers *Phil. Mag.* 2 20–30
- Haidinger W. 1849 Ueber die schwarzen und gelben Parallel-Linien am Glimmer *Poggendorff's Annal.* (Lpz) LXXVII 219–228
- Haidinger W. 1855 Die Interferenzlinien am Glimmer. Berührungsringe und Plattenringe *Poggendorff's Annal.* (Lpz) XCVI 453–468
- Boulouch M. R. 1893 Dédoublément des franges d'interférence en lumière naturelle *J. Physique* 2 316–320
- Fabry Ch. and Perot, A. 1897 Sur les franges des lames minces argentées et leur application à la mesure de petites épaisseurs d'air *Ann. Chim. Phys.* 12 459–501
- Edser E. and Butler C. P. 1898 A simple method of reducing prismatic spectra *Phil. Mag.* 46 207–216
- Perot A. and Fabry Ch. 1899a Sur l'application de phénomènes d'interférence à la solution de divers problèmes de spectroscopie et de métrologie *Bull. Astronomique* 16 5–32 (Engl. *Transi. Astrophys. J* 9 87–115)
- Fabry Ch. and Perot A. 1899 Théorie et applications d'une nouvelle méthode de spectroscopie interférentielle *Ann. Chim. Phys.* 16 115–144
- Perot A. and Fabry Ch. 1899b Méthodes interférentielles pour la mesure des grandes épaisseurs et la comparaison des longueurs d'onde *Ann. Chim. Phys.* 16 289–338
- Fabry Ch. , Macé de Lépinay J. and Perot A. 1899a Mesure en longueurs d'onde d'un cube de quartz de 4 cm de côté *C. R Acad. Sci., Paris* 128
- Fabry Ch. , Macé de Lépinay J. and Perot A. 1899b Sur la masse du décimètre cube d'eau *C R. Acad. Sci., Paris* 129
- Fabry Ch. and Perot, A. 1900a Sur les sources de lumière monochromatiques *J. Physique* 9 369–382
- Fabry Ch. and Perot, A. 1900b Nouvelle source de lumière pour la spectrométrie de précision *C R. Acad. Sci., Paris* 130 406–409
- Perot A. and Fabry Ch. 1900 Détermination de nouveaux points de repère dans le spectre *C R. Acad. Sci., Paris* 130 492–495
- Fabry Ch. and Perot, A. 1901 Sur un nouveau modèle d'interféromètre *Ann. Chim. Phys.* 22 (Engl. *Transi. Astrophys. J* 13 265–72, PLATE IX)
- Perot A. and Fabry Ch. 1901 Mesure en longueurs d'onde de quelques étalons de longueur a bouts *Ann. Chim. Phys.* 24 119–139
- Fabry Ch. and Perot, A. 1902 Mesures de longueurs d'onde en valeur absolue, spectre solaire et spectre du fer *Ann. Chim. Phys.* 25 (Engl. *Transi. Astrophys. J* 15 73–96, 261–73)
- Lummer O. and Gehrke L. 1903 Ueber die Anwendung der Interferenzen an planparallelen Platten zur Analyse feinsten Spektrallinien *Drude's Annalen* X 457
- Perot A. and Fabry Ch. 1904 Sur la séparation des raies spectrales très voisines a propos d'un travail récent de MM. Lummer et Gehrke *J. Physique* 3 28–32
- Barnes J. 1904 On the analysis of bright spectrum lines *Astrophys. J.* 19 190–211
- Fabry Ch. 1904 On the wavelength of the cadmium line at $\lambda 5086$ *Astrophys. J.* 19 116–118
- Fabry Ch. and Perot, A. 1904 On the corrections to Rowland's wavelengths *Astrophys. J.* 19 119–120
- Fabry Ch. 1905 Sur un nouveau dispositif pour l'emploi des méthodes de spectroscopie interférentielle *C R. Acad. Sci., Paris* 140 848–851
- Hamy M. 1906 Sur les franges de réflexion des lames argentées *J. Physique* 5 789
- Lord Rayleigh 1906a Some measurements of wave-lengths with a modified apparatus *Phil. Mag.* 11 685–703
- Lord Rayleigh 1906b On the interference-rings, described by Haidinger, observable by means of plates whose surfaces are absolutely parallel *Phil. Mag.* 12 489–493
- Eversheim P. 1907 Determination of wave-lengths of light for the establishment of a standard system *Astrophys. J.* 26 172–190 (translated from *Z fur Wissensch. Photographie* 5 152–180)
- Lord Rayleigh 1908 Further measurements of wave-lengths and miscellaneous notes on Fabry and Perot's apparatus *Phil. Mag.* 15 548–558
- von Baeyer O. 1908 Über den Einfluss der Dispersion auf den Gangunterschied beim Interferenzspektroskop nach Lummer-Gehrcke *Phys. Zeitschr.* 9 831–834
- Fabry Ch. and Buisson, H. 1908 (translated by R R Tatnall) Wave-length measurements for the establishment of a system of spectroscopic standards *Astrophys. J.* 28 169–196
- Pfund A. H. 1908 A redetermination of the wave-lengths of standard iron lines *Astrophys. J.* 28 197–211
- Fabry Ch. and Buisson, H. 1909 Comparaison des raies du spectre de l'arc électrique et du soleil. Pression de la couche renversante de l'atmosphère solaire *C. R Acad. Sci., Paris* 148 688–690
- Fabry Ch. and Buisson, H. 1910a Interférences produites par les raies noires du spectre solaire *J. Physique* 9 197–205
- Fabry Ch. and Buisson, H. 1910b Application de la méthode interférentielle à la mesure de très petits déplacements de raies. Comparaison du spectre solaire avec le spectre d'arc du fer. Comparaison du centre et du bord du soleil *Astrophys. J.* 31 97–119
- Buisson H. and Fabry Ch. 1910 Mesures de petites variations de longueurs d'onde par la méthode interférentielle. Application à différents problèmes de spectroscopie solaire *J. Physique* 9 298–316
- Eversheim P. 1910 Measurements of wave-lengths of standard iron lines *Astrophys. J.* 31 76–77
- Perot A. and Mlle Lindstedt 1911 Sur la longueur d'onde de la raie solaire b_2 *C. R Acad. Sci., Paris* 152 1367–1370
- Fabry Ch. and Buisson, H. 1911 Application of the interference method to the study of nebulae *Astrophys. J.* 33 406–409
- Goos F. 1912 Eine neue Form des Fabry-Perotschen Interferometers mit veränderlichem Plattenabstand *Z Instrum.* 32 326–328
- Perot A. 1912 Sur le mouvement apparent des vapeurs dans l'atmosphère solaire *C R. Acad. Sci., Paris* 154 1684–1686
- Buisson H. and Fabry Ch. 1912 La largeur des raies spectrales et la théorie cinétique des gaz *J. Physique* 2 442–464
- Benoit J. R. , Fabry C. and Perot A. 1913 Nouvelle détermination du rapport des longueurs d'onde fondamentales avec l'unité metrique *Trav. Bur. Int. Pds. Mes.* 15 3
- Buisson H. and Fabry Ch. 1913 Sur les longueurs d'onde des raies du krypton *C. R Acad. Sci., Paris* 156 945–947
- Fabry Ch. and Buisson, H. 1914 Vérification expérimentale du principe de Doppler - Fizeau *C R. Acad. Sci., Paris* 158 1498–1499
- Buisson H. , Fabry Ch. and Bourget, H. 1914 An application of interference to the study of the orion nebula *Astrophys. J.* 40 241–258
- Merton T. R. 1915 On the application of interference methods to the study of the origin of certain spectrum lines *Proc. R Soc. A* 91 421–431
- Lord Rayleigh 1915 On the widening of spectrum lines *Phil. Mag.* 29 274–284

- Fabry Ch. and Buisson, H. 1919 Description et emploi d'un nouveau microphotomètre J. Physique 9 37–46
- Merton T. R. 1920 On the secondary spectrum of hydrogen Proc. R Soc. A 96 382–388; On the spectra of isotopes Proc. R Soc. A 96 388–395
- Buisson H. and Fabry Ch. 1921 Sur le déplacement des raies solaires sous l'action du champ de gravitation C. R Acad. Sci., Paris 172 1020–1022
- Perot A. 1921 Mesure de la pression de l'atmosphère solaire dans la couche du magnésium et vérification du principe de relativité C. R Acad. Sci., Paris 172 578–581
- St John Ch. E. and Babcock H. D. 1921 Wave lengths of lines in the iron arc from grating and interferometer measures $\lambda 3370$ – $\lambda 6750$ Astrophys. J. 53 260–299
- Fabry Ch. 1923 Les Applications des Interférences Lumineuses Éditions de la Revue d'Optique théorique et instrumentale (Paris)
- Feussner W. and Janicki L. 1927 Interferenzerscheinungen dünner Blattchen: 10. Interferometer von Fabry und Perot in Handbuch der Physikalischen Optik ed. E. Gehrcke (Leipzig: Barth) pp 430–433
- Kösters W. 1927 Anwendung der Interferenzen zu Messwecken in Handbuch der Physikalischen Optik ed. E. Gehrcke (Leipzig: Barth) pp 476–485